Carbon Action Plan Independent Study (2026)

# Modeling Cooper Union Energy Efficiency Opportunities

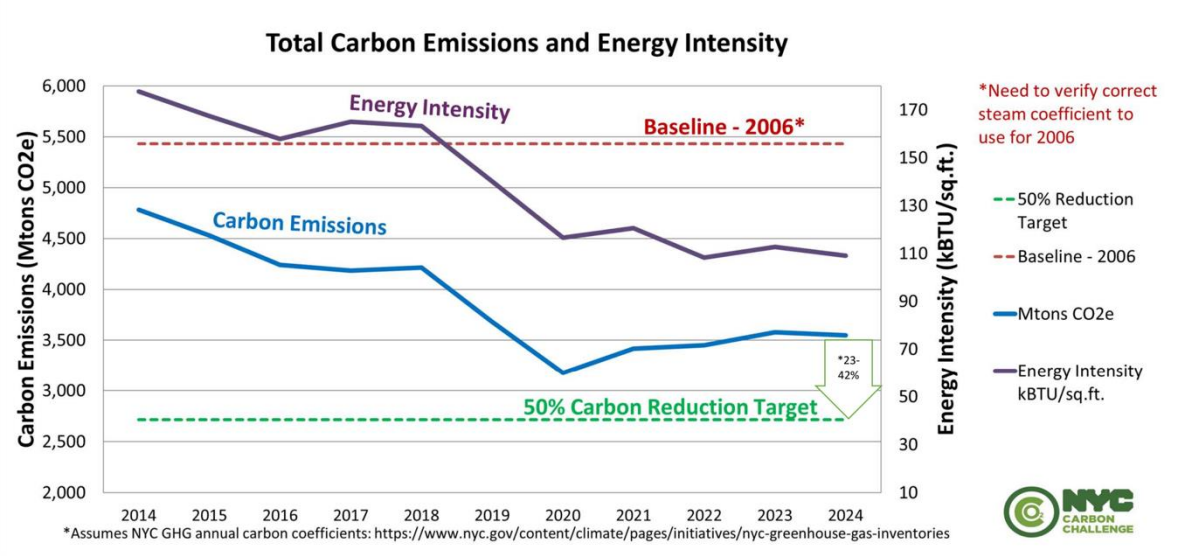
**Ayushi Bhattacharjee**
**Danielle Chiu**
**Hyungkyu (Tom) Lee**
**Jhanjaneth Perea**



# Problem and Scope

**GOAL**: Currently for 2026, reduce carbon emissions **50%** from **2006**-2030.
Previously in 2017, reduce carbon emissions **40%** from **2014**-2030.

This is based on the NYC Carbon Challenge and LL97.




Source: CU Climate Action Plan (CAP) Executive Summary - October 2025

# Cooper Campus Wide Emissions

$tCO_2e$ based on 2025 updated annual NYC CC coefficients

**2024 Total Emissions**

3,541 $tCO_2e$

**Contribution by Source**

66.5% Electricity

33.5% Natural Gas

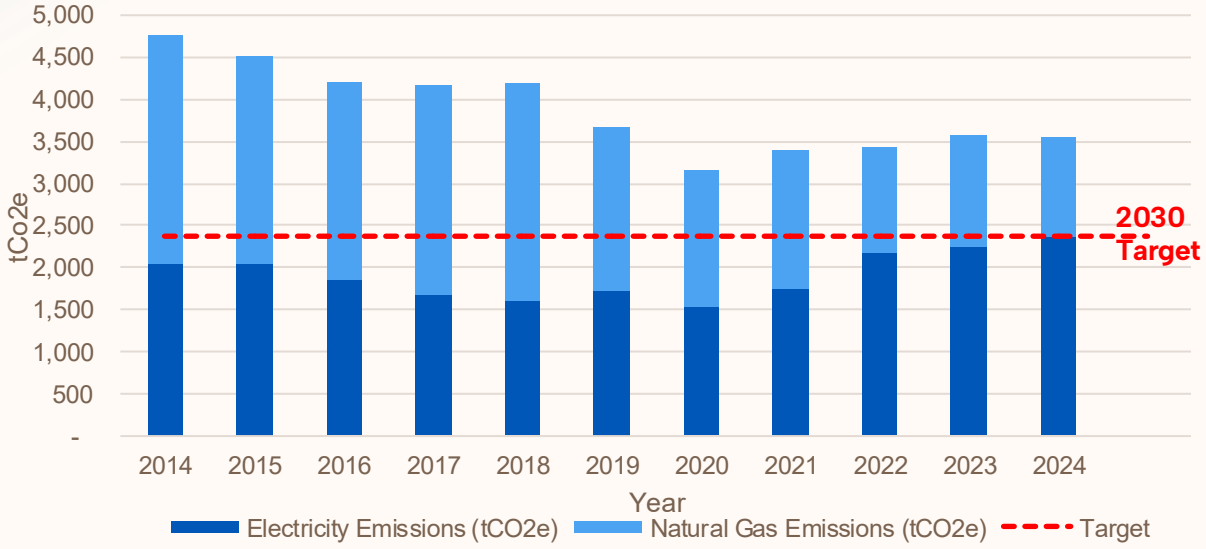
**Contribution by Building**

58% 41Cooper Square

32% Foundation Building

10% Residence Hall

Campus Total Emissions by Source Trends ($tCO_2e$)




**Our Commitment**

50 % reduction in Emissions by 2030
– NYC Carbon Challenge

**Goal:** Reach

2,367 $tCO_2e$

by 2030

**NYC Climate Mobilization Act**

Local Law 97 Requires buildings in NYC over 25,000 sqft to reach zero net carbon emissions by 2050

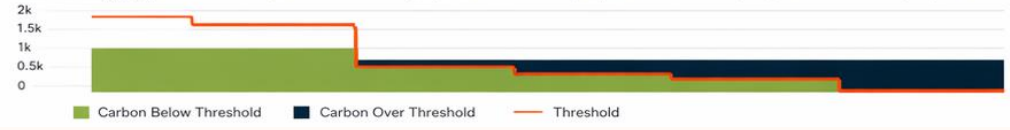
# All Cooper Buildings are predicted to face fines in the 2030s due to Local Law 97 penalties.

1 **Foundation Building**

Old School control and efficiency measures

**Annual Carbon Threshold Summary**

| $tCO_2e$/yr | 2024–2029 | 2030–2034 | 2035–2039 | 2040–2049 | 2050+ |
|---|---|---|---|---|---|
| Emissions | 989 | 655 | 655 | 655 | 655 |
| **Threshold** | 1,877 | 399 | 235 | 34 | 0 |
| **Est. Penalty ($/yr)** | $0 | $68,610 | $112,608 | $166,395 | $175,610 |



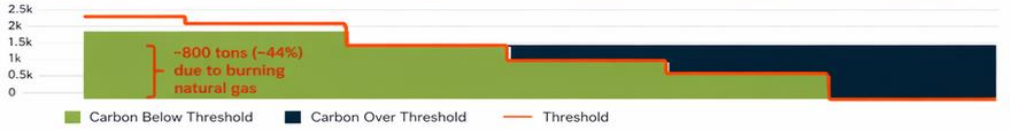

$68,610 / yr
LL97 penalties expected by 2030

2 **41 Cooper Square**

High Natural Gas: Electricity emissions

**Annual Carbon Threshold Summary**

| $tCO_2e$/yr | 2024–2029 | 2030–2034 | 2035–2039 | 2040–2049 | 2050+ |
|---|---|---|---|---|---|
| Emissions | 1,839 | 1,322 | 1,322 | 1,322 | 1,322 |
| **Threshold** | 2,335 | 1,387 | 995 | 577 | 0 |
| **Est. Penalty ($/yr)** | $0 | $0 | $87,661 | $199,674 | $354,310 |

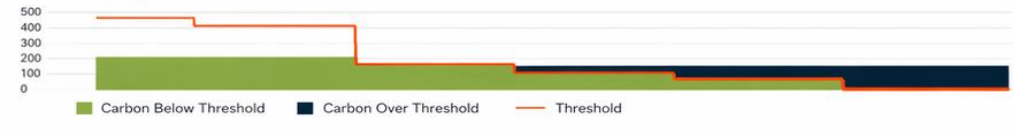



$87,661 / yr
LL97 penalties expected by 2035

3 **Residence Hall**

**Annual Carbon Threshold Summary**

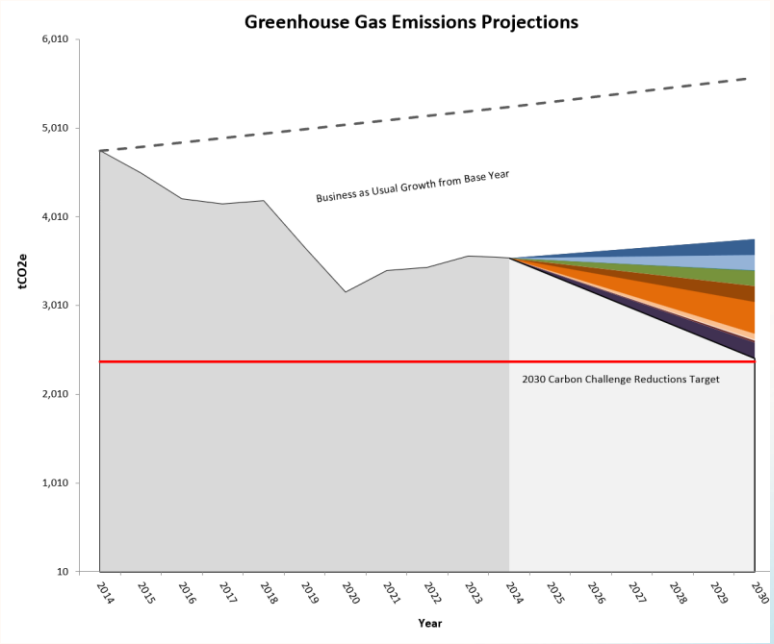
| $tCO_2e$/yr | 2024–2029 | 2030–2034 | 2035–2039 | 2040–2049 | 2050+ |
|---|---|---|---|---|---|
| Emissions | 247 | 165 | 165 | 165 | 165 |
| **Threshold** | 465 | 161 | 96 | 46 | 0 |
| **Est. Penalty ($/yr)** | $0 | $885 | $18,531 | $31,833 | $44,138 |




$885 / yr
LL97 penalties expected by 2035

# HVAC optimization is needed to achieve the 2030 Carbon Challenge Reductions target and avoid Local Law 97 penalties.

- Retrofitting Opportunities
  - HVAC Optimization – In Progress
    Capital Upgrades – Under Evaluation
  - LED Replacement – Done
- Large drop from 2014-2024
- Slight increase/plateau after 2020



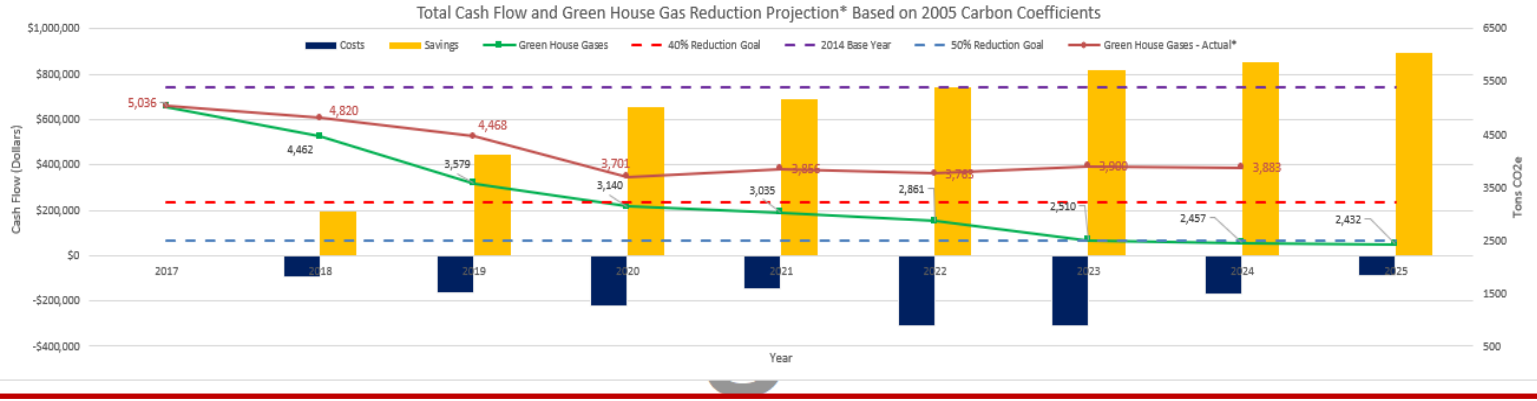
**The Cooper Union - New York, NY - Energy Master Plan**

**Project Information**

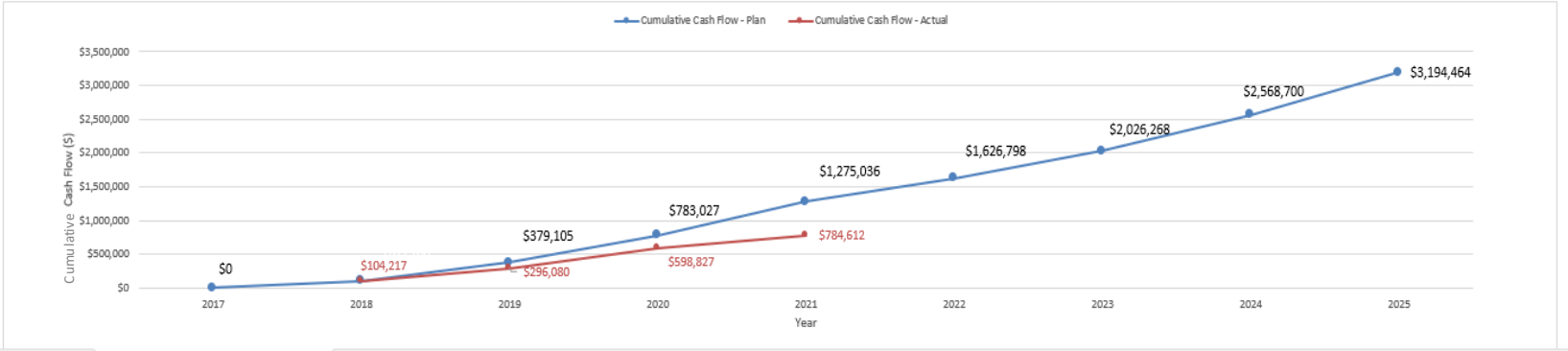
**Project Location:** Cooper Union, New York, NY
**Project Title:** Cooper Union Energy Master Plan
**Project Engineer:**
**Date:** Wednesday, June 5, 2019

**Capital Budget Forecast**

| Year | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|---|
| Proposed Projects | 5 | 14 | 2 | 1 | 2 | 3 | 3 | 1 |
| Total CapEx | $91,661 | $164,509 | $223,232 | $148,741 | $311,476 | $306,996 | $171,236 | $90,000 |
| Urgency | Immediately | 1 Year | 1 to 2 Years | 3 to 5 Years | 3 to 5 Years | 5+ Years | 5+ Years | 5+ Years |




These values are estimated based on the 2021 cost of energy




Source: CU Climate Action Plan (CAP) Executive Summary - October 2025

These values are estimated based on the 2021 cost of energy

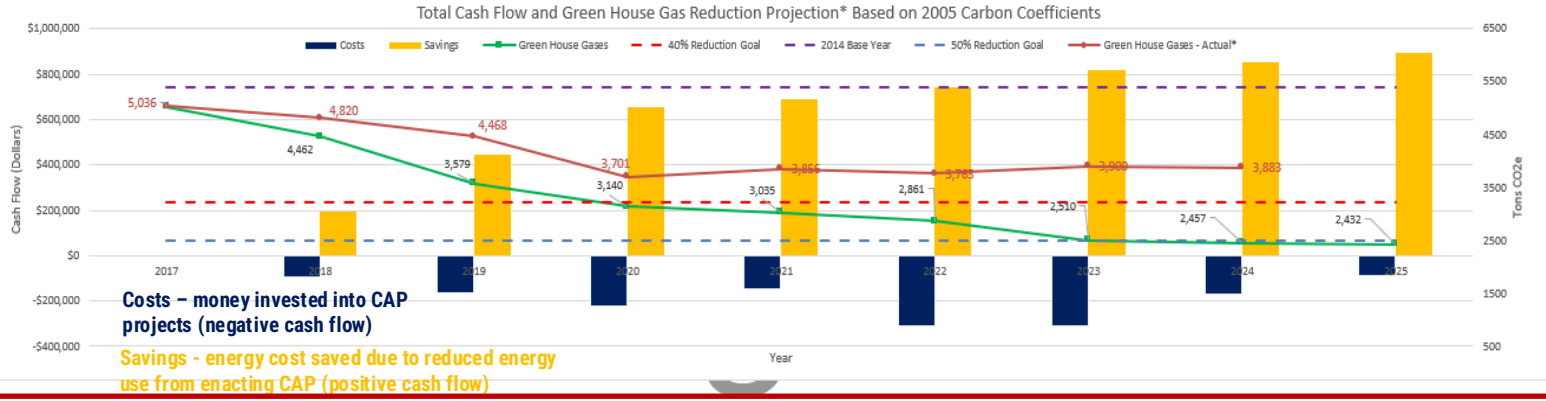
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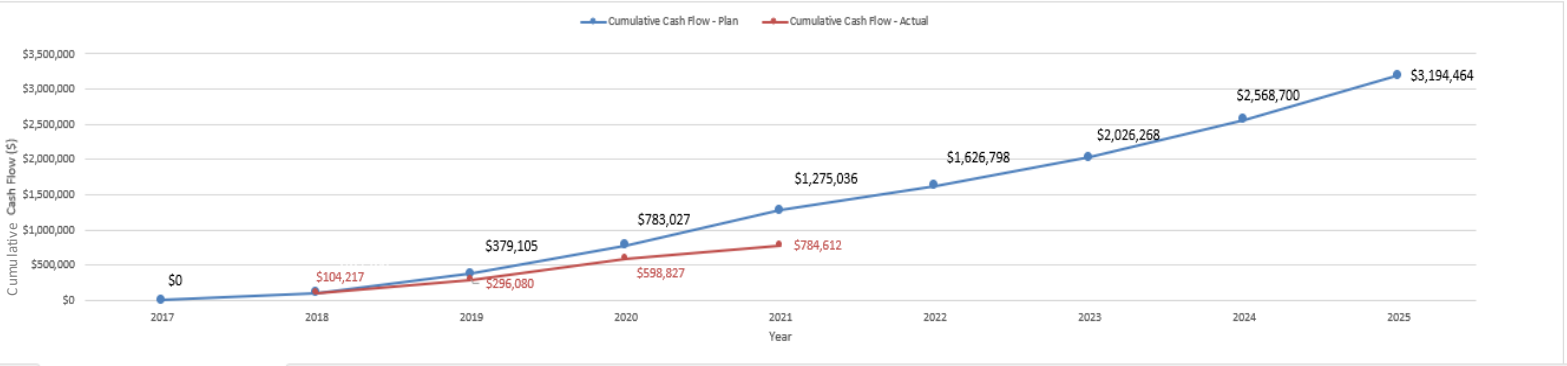
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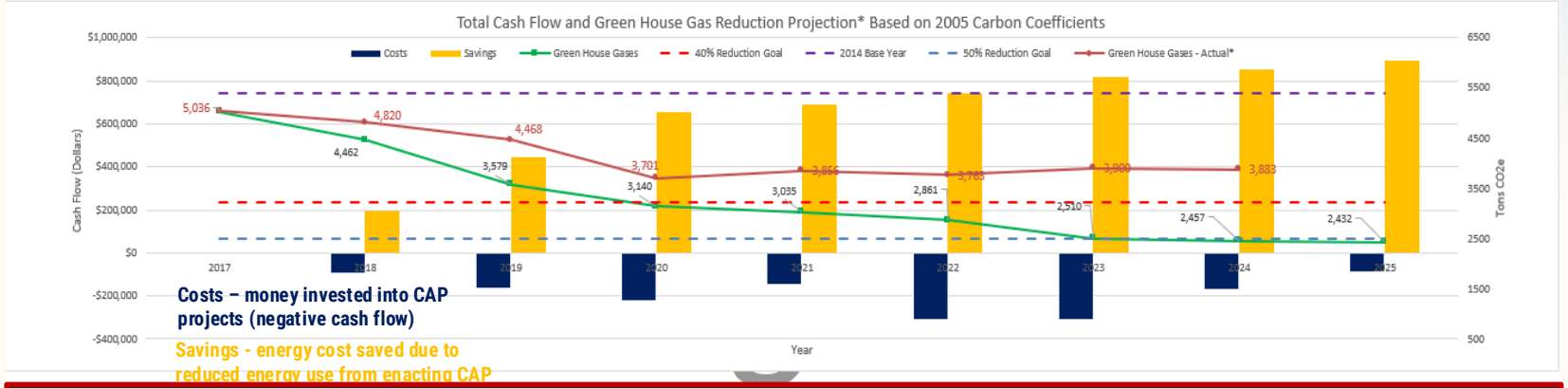
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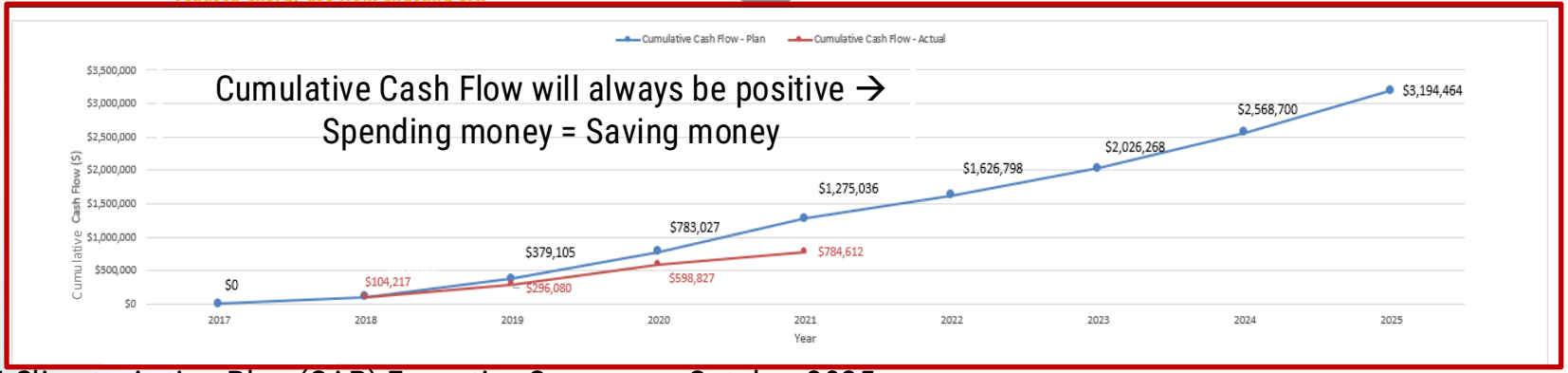
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Combining the costs and savings to get the cumulative cash flow…




Source: CU Climate Action Plan (CAP) Executive Summary - October 2025

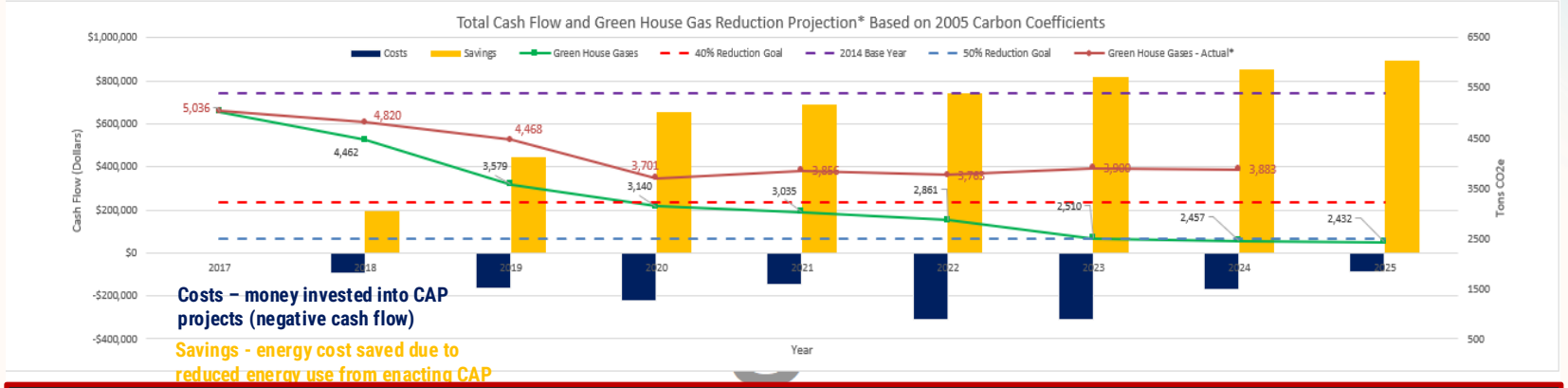
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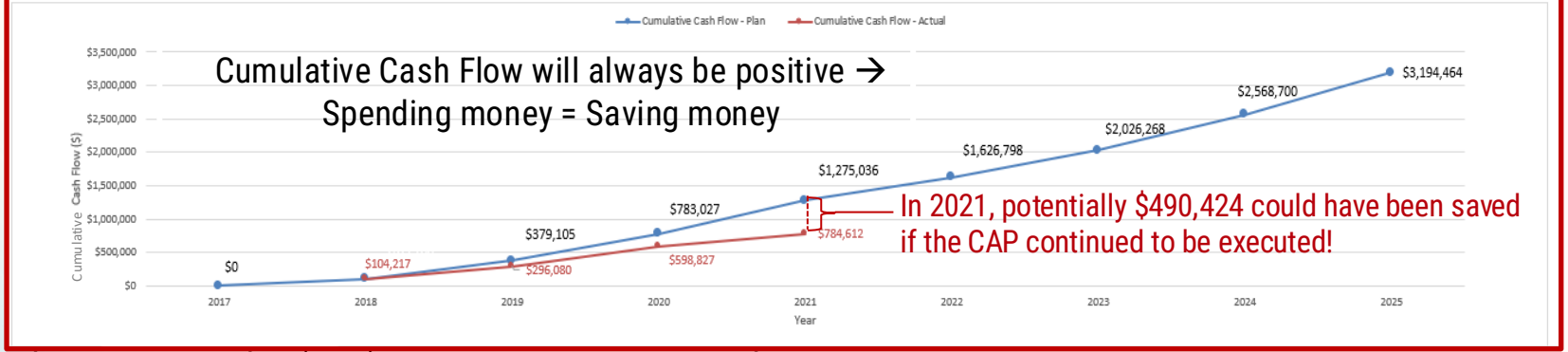
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Combining the costs and savings to get the cumulative cash flow…




Source: CU Climate Action Plan (CAP) Executive Summary - October 2025

# Past CAP Projects and tCO2e Savings

- Lab ACH Optimization and Min Flow Resets (344 tCO2e)
- AHU Discharge Air Temperature (DAT) Optimization (247 tCO2e)
- Damper Actuator Repairs (344 tCO2e)
- LED Lighting Retrofit (184 tCO2e)

- **We were predicted to save $3 million and 2,500 tCO2e by 2025**
- Cooper achieved some of it, but there is still progress to be made
- **In this presentation we revisit the top 2 Energy Cost Measures (ECMs)**
  - **41CS HVAC Optimization**
  - **Foundation Building HVAC Optimization**

# Savings from this Semester's Work

Incorporating this semester's work could potentially save…

**Cooper Union Carbon Action Plan - ECM Summary**

| Identification | | | | Baseline (Annual) | | | | Proposed (Annual) | | | | Savings (Annual) | | | | Capital and Payback (ROM) | | | | Notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Building | ECM | ECM Description | Model Includes | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Capital Expense (ROM) | Potential Rebate (ROM) | After Rebate Expense (ROM) | Simple Payback | |
| | | | | *kWh* | *Therms* | *$* | *MTCO2e* | *kWh* | *Therms* | *$* | *MTCO2e* | *kWh* | *Therms* | *$* | *MTCO2e* | *$* | *$* | *$* | *Years* | |
| 41CS | ECM-1A | Night Setback (AHU-3/6 Off):<br>AHU-3/6 off at night (10 PM to 7 AM) | AHU-3/6 | 731586 | 63966 | 217791 | 551 | 461729 | 39407 | 136312 | 343 | 269,857 | 24,559 | $81,479 | 208.4 | $2,000 | - | $2,000 | 0.1 yrs | BMS schedule change. AHU-3 and AHU-6 off at night. |
| 41CS | ECM-1B | Night Setback (Aggressive):<br>AHU-3/6 reduce by 50% at night<br>AHU-4/5 reduce by 24% CFM at night | AHU-3/4/5/6 | 1750155 | 145795 | 512443 | 1280 | 1540658 | 133190 | 456850 | 1153 | 209,497 | 12,605 | $55,592 | 127.5 | $2,000 | - | $2,000 | - | Aggressive nighttime CFM reduction on AHU-3/4/5/6. Programming only. |
| 41CS | ECM-1C | Night Setback (Moderate):<br>AHU-3/6 reduce by 20% at night<br>AHU-4/5 reduce by 12% CFM at night | AHU-3/4/5/6 | 1750155 | 145795 | 512443 | 1280 | 1651463 | 141166 | 464883 | 1227 | 98,692 | 4,629 | $47,560 | 53.1 | $2,000 | - | $2,000 | - | Moderate nighttime CFM reduction on AHU-3/4/5/6. Programming only. |
| 41CS | ECM-2 | Zone Level Optimization:<br>AHU-3/6: 3,000 to 5,000 CFM reductions<br>AHU-4/5: 13,000 CFM reduction | AHU-3/4/5/6 | 1720485 | 145041 | 505793 | 1267 | 1289782 | 108645 | 379066 | 950 | 430,704 | 36,396 | $126,727 | 317.8 | $30,000 | - | $30,000 | - | Zone-level controls repair (failed valves, dampers, sensors) and setpoint optimization. |
| 41CS | ECM-3 | Resolve AHU-XX Pressure Drop | AHU-3 | 363531 | 34197 | 111082 | 287 | 291437 | 34197 | 97096 | 266 | 72,094 | - | $13,986 | 20.8 | $2,000 | - | $2,000 | - | Investigate root cause of pressure drop. Scope and cost TBD. |
| FB | ECM-1A | HVAC Scheduling:<br>Turn off AHUs 12 AM to 5 AM | Whole Building | | | | | 2,466,900 | 57,300 | $546,536 | 1,017.2 | 253,200 | 19,400 | $72,129 | 176.2 | $60,000 | | $60,000 | 0.8 yrs | Stand-alone time clock at each of 17 non-Siemens AHUs, interrupts fan starter at MCC. |
| FB | ECM-1B | Night Space Temp Setbacks:<br>Set back space temp setpoints 10 PM to 7 AM (60 / 85 F) | Whole Building | | | | | 2,675,900 | 65,000 | $596,215 | 1,118.4 | 44,200 | 11,700 | $22,451 | 74.9 | $600,000 | | $600,000 | 26.7 yrs | Full DDC retrofit at 17 non-Siemens AHUs: controllers, sensors, valves, damper actuators, BMS network. |
| FB | ECM 1A + 1B | HVAC Scheduling:<br>Turn off AHUs 12 AM to 5 AM<br>+ Setback space temp setpoints 10 PM to 12 AM | Whole Building | 2,720,100 | 76,700 | $618,666 | 1,193.4 | 2,429,800 | 54,700 | $536,255 | 992.6 | 290,300 | 22,000 | $82,410 | 200.7 | $600,000 | | $600,000 | 7.3 yrs | Full DDC retrofit at 17 non-Siemens AHUs: controllers, sensors, valves, damper actuators, BMS network. |
| FB | ECM-2 | VAV Operation:<br>Operate VFD AHUs as variable volume (currently constant volume) | Whole Building | | | | | 2,649,800 | 78,100 | $606,688 | 1,180.5 | 70,300 | (1,400) | $11,978 | 12.9 | - | | - | - | Investigate and recommission 9 Carrier VFD VAV units to restore static pressure reset. |
| FB | ECM-3 | Convert MZ to VAV:<br>Convert multizone AHUs to variable volume | Whole Building | | | | | 2,503,200 | 72,700 | $571,843 | 1,109.4 | 216,900 | 4,000 | $46,823 | 83.9 | $1,000,000 | | $1,000,000 | 21.4 yrs | MZ-to-VAV retrofit on 12 multizone AHUs: VFDs, AFMAs, full DDC, zone damper separation. |

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# Foundation Building

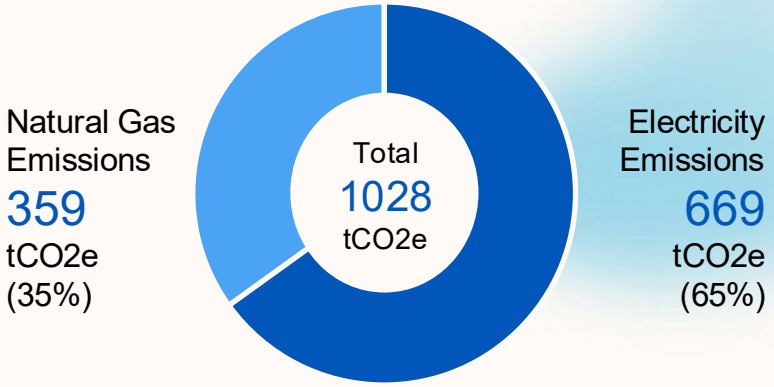
# Cooper Union Foundation Building

## BULIDING SNAPSHOT



- Built in 1859
- Today designated as a New York City landmark
- Total building area: 190,144 sq ft

## EMISSIONS BY SOURCE IN 2025



# Foundation Building Emissions

$tCO_2e$ based on LL 97 Carbon CC

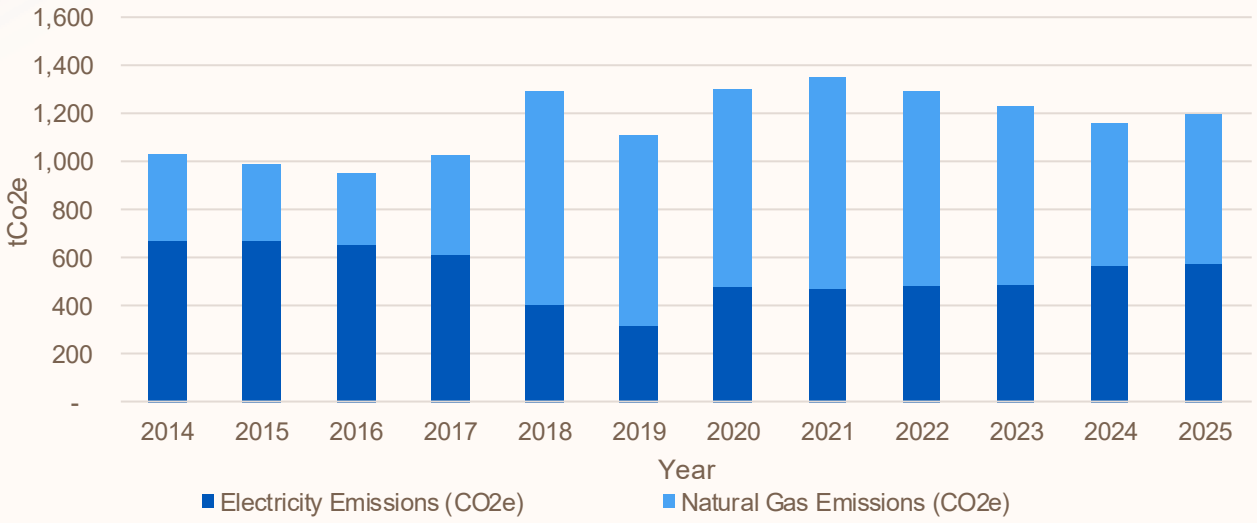
2025 Total Emissions

1,028 $tCO_2e$

2025Emissions Intensity (EUI)

11.92 $tCO_2e$ / sq ft

Campus Total Emissions by Source Trends ($tCO_2e$)




**Goal:** Reach

399 $tCO_2e$

by 2030

235 $tCO_2e$

by 2035

34 $tCO_2e$

by 2040

0 $tCO_2e$

by 2050

# Foundation Building Inventory

**Total Number of AHU**: 30 traced units

**AHUs on BMS**: 11

**AHU Types**: Single Zone, Multizone, Constant Volume, Variable Air Volume

| Unit | Unit Audit Report | Location | Service | On BMS | Make | Air Flow | CV/VF |
|---|---|---|---|---|---|---|---|
| | | | | | | Test CFM | CAV/VAV |
| MZ-1 | MZ-1 | Roof (800) | 1) Peter Cooper Suite (800, 801);<br>2) Exhibition Area, Offices (700, 701, 703 704);<br>3) Office (702) | No | McQuay | 12,612 | CV CAV |
| MZ-2 | MZ-2 | MER (721) | 1) Corridor (720);<br>2) Rm 712;<br>3) Rm 713;<br>4) Rm 714;<br>5) Rm 715 | No | McQuay | 2,850 | CV CAV |
| MZ-3 | MZ-3 | MER(621) | 1) Rm 621;<br>2) Rm 613;<br>3) Rm 614;<br>4) Rm 615; | No | McQuay | 2,920 | CV CAV |
| MZ-4 | MZ-4 | MER(632) | 1) Rm 629;<br>2) Rm 630;<br>3) Rm 631;<br>4) Rm 632; | No | McQuay | 15,354 | CV CAV |
| MZ-5 | MZ-5 | MER(534) | 1) Corridor (521)<br>2) Corridor (520), Storage (534);<br>3) Silk Screen Shop NE (544);<br>4) Silk Screen Shop NW (544);<br>5) Classroom (537);<br>6) Classroom (543); | No | McQuay | 8,862 | CV CAV |
| MZ-6 | MZ-6 | MER(534) | 1) Classroom (515);<br>2) Graphisics Shops (514, 514A);<br>3) Graphics Shops (512, 513); | No | McQuay | 5,958 | CV CAV |
| MZ-7 | Mz-7 | MER(534A) | 1) Photo Studio NE (530);<br>2) Photo Studio SE (530);<br>3) Classroom (539);<br>4) Classroom (541);<br>5) Projection Booth (538);<br>6) Projection Booth (540); | No | McQuay | 13,005 | CV CAV |
| MZ-8 | MZ-8 | MER(534) | 1) Exhibition Area (600);<br>2) Office (601);<br>3) Exhibit Area (500);<br>4) Exhibit Area (501); | No | McQuay | 11,059 | CV CAV |
| SZ-8 | SZ-8 | Office (213A) | Exhibition Area, Corridor (Rm 230, 220) | No | McQuay | 4,201 | CV CAV |
| MA-2 | MA-2 | F15MER | Make-up Air for 5th Floor Paint Shop | No | Trane | 1,902 | CV CAV |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| AH-4A | AH-4A | Woodwork Shop (430) | VAV-401: Woodwork Shop (430);<br>VAV-402(RHC): Hi Vent Paint / Plaster Rm 9421);<br>VAV-403: 3D Design Studio (415);<br>VAV-404: Sculpture Studio (414);<br>VAV-405: Sculpture Studio (413);<br>VAV-406: Plastic Rm (412) | Yes | Carrier | 15,293 | VFD VAV |
| AH-4B | AH-4B | Woodwork Shop (430) | VAV-407: Woodwork Shop (430);<br>VAV-408: Woodwork Shop (430); | Yes | Carrier | 2,631 | VFD VAV |
| AH-4C | AH-4C | Woodwork Shop (430) | VAV-409: Sculpture Studio (401A);<br>VAV-410: Sculpture Studio (400); | Yes | Carrier | 496 | VFD VAV |
| AH-3A | AH-3F | Office (338) | VAV-309-thru 314: Architecture Studio (330) | Yes | Carrier | 5,270 | VFD VAV |
| AH-3B | AH-3B | Office (337) | VAV-315: Office (338);<br>VAV-316: Student Workspace (313);<br>VAV-317: Student Workspace (314);<br>VAV-318: Student Workspace (312); | Yes | Carrier | 2,359 | VFD VAV |
| AH-3D | AH-3D | Office (335) | VAV-301-thru 308: Architecture Studio (330) | Yes | Carrier | 5,512 | VFD VAV |
| AH-3C | AH-3C | Office (332) | 1) Lobby Area (300);<br>2) Admin Offices (301); | Yes | Carrier | 4,569 | VFD VAV |
| AH-3F | AH-3F | Corridor (320) | VAV (w/RH): Lecture Hall (315) | Yes | Carrier | 3,870 | VFD VAV |
| MZ-18 | MZ-18 | MER (212A) | 1) Classroom (215); 2) Office (214) | No | Carrier | 3,644 | CV CAV |
| MZ-19 | MZ-18 | MER (212A) | 1) Office (212); 2) Office (213) | No | Carrier | 3,715 | CV CAV |
| MZ-20 | | Mezz MER | 1) Library Workroom; 2) Mezz Corr; 3)FL 2 Corr | No | Carrier | N/A | CV CAV |
| MZ-21 | | Mezz MER | 1) A/V Center; 2) Seminar Rm; 3) Library Workroom | No | Carrier | N/A | CV CAV |
| AH-3E | | Corridor (320) | VAV (w/ RH): Lecture Hall (315) | Yes | Carrier | N/A | VFD VAV |
| SZ-1 | AHU 1 | Library Corridor (135) | Main Library (Rm 130) | No | Carrier | 1,435 | CV CAV |
| SZ-2 | AHU 2 | Library Corridor (135) | Library Corridor (Rm 135) | No | Carrier | 2,134 | CV CAV |
| SZ-3 | AHU 3 | Lobby MER (202A) | Lobby/Elevators (Rm 100) | No | Carrier | 2,699 | CV CAV |
| SZ-4 | SZ-4 | Cellar MER (B13) | Stacks (Rm 110-115) | No | McQuay | 6,000 | CV CAV |
| SZ-5 | SZ-5 | Sub-Cellar MER (B18) | Great Hall NE (RM B24) | Yes | McQuay | 5,567 | CV CAV |
| SZ-6 | SZ-6 | Sub-Cellar MER (B18) | Great Hall SW, Lobby, Stairwell (RM B24, B00, B07) (NOT OPERATIONAL) | Yes | McQuay | 3,988 | CV CAV |
| SZ-7 | SZ-7 | Classroom (214A) | Offices, Conidor (RM 237-243) | No | McQuay | 2,500 | CV CAV |
| SZ-8 | SZ-8 | Office (213A) | Exhibition Area, Corridor (Rm 230, 220) | No | McQuay | 4,201 | CV CAV |
| MA-2 | MA-2 | F15MER | Make-up Air for 5th Floor Paint Shop | No | Trane | 1,902 | CV CAV |

# Foundation Building 2018 Audit

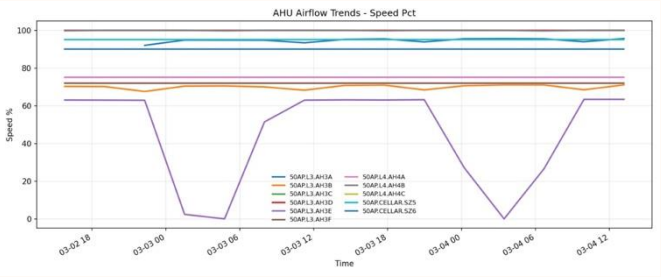
| ECM | | Electricity Reduction | | Gas Reduction | Energy Reduction | $CO_2$ Reduction | Cost of implementation | | | Measure Savings | Payback |
|---|---|---|---|---|---|---|---|---|---|---|---|
| No. | Description | Usage kWh | Demand kW | Therms | MMBtu | Tons | First Cost | Rebate | Net Capex After Rebate | Blended | Simple |
| ECM.1 | AHU Night Shutoff Controls, VFDs & Motor Replacement | 669,925 | 4.3 | - | 2,285.88 | 283.18 | $334,987 | $91,969 | $243,019 | $83,852 | 3.99 |
| ECM.1a | AHU replacement w/ VFDs | 669,925 | 4.3 | - | 2,285.88 | 283.18 | $1,664,676 | $109,770 | $1,554,906 | $83,852 | 19.85 |
| ECM.1b | AHU Timeclocks | 386,401 | - | - | 1,318.46 | 163.33 | $52,800 | $26,400 | $26,400 | $47,561 | 1.11 |
| ECM.1c | BMS Expansion | 736,917 | 4.73 | - | 2,514.47 | 311.5 | $5,000,000 | $120,747 | $4,879,253 | $92,237 | 54.21 |
| ECM.2 | Perimeter HHW Radiator Valves, VFDs & Motor Replacement | 35,408 | 0.06 | 19,036 | 2,024.44 | 116.18 | $159,435 | $59,979 | $99,456 | $21,035 | 7.58 |
| ECM.3 | Fix Absorber | 128,956 | 60.78 | - | 440.02 | 54.51 | $10,000 | $10,000 | $- | $35,550 | 0.28 |
| ECM.4 | Remove Throttle from HHWP Discharge Valves | 7,162 | 1.55 | - | 24.44 | 3.03 | $- | $- | $- | $1,383 | - |
| ECM.5 | Remove Throttle from CHWP Discharge Valves | 3,960 | 2.79 | - | 13.51 | 1.67 | $- | $- | $- | $1,391 | - |
| ECM.6 | Solar PV Array | 54,854 | - | - | 187.17 | - | $78,320 | $- | $78,320 | $6,752 | 11.6 |
| ECM.7 | Repair SZ-6 | 23,182 | 9.21 | - | 79.1 | 9.8 | $44,597 | $9,234 | $35,363 | $5,835 | 7.64 |
| **Total** | | **923,445** | **78.69** | **19,036** | **5,054.56** | **468.37** | **$627,339** | **$171,182** | **$456,157** | **$155,797** | **4.03** |
| **% Savings** | | **54.60%** | **17.30%** | **12.60%** | **24.70%** | **31.90%** | **-** | **-** | **-** | **33.60%** | **-** |

- Previous audit found greatest savings in AHUs
- ECMs reducing 100-300 tons of CO2 and saving $40,000-$90,000

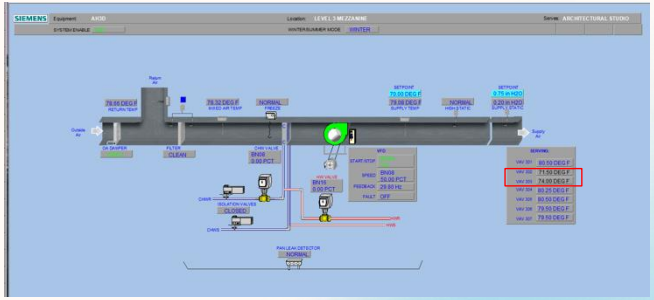
# Field Observations: BMS and FB Walkthrough



Foundation Building Walkthrough




Dynamic BMS Airflow Trend



BMS Screenshot with Locked VAV

SZ-4
- Singlezone
- No actuators
- Variable duct

SZ-1 and SZ-2
- No actuator
- Outside mixes with return air

MZ-3, MZ-4, MZ-5, MZ-6
- CV and in ceiling
- MZ-5 and MZ-6 replaced with new units

MZ-7 and MZ-8
- Have not been replaced
- Both CV

2 New Units
- MA-2 make up unit, 521 papermaking room
- No VFD
- 100% outside air

AH3D, AH3B
- VAVs not operating correctly, stuck as CV

- Verified AHU types: Multizone, Singlezone, Constant Volume AHUs, Variable Volume AHUs
- Multizone damper configuration: conventional
- No BMS historical data available
- AHU treated as running 24/7

# Foundation Building eQuest Model

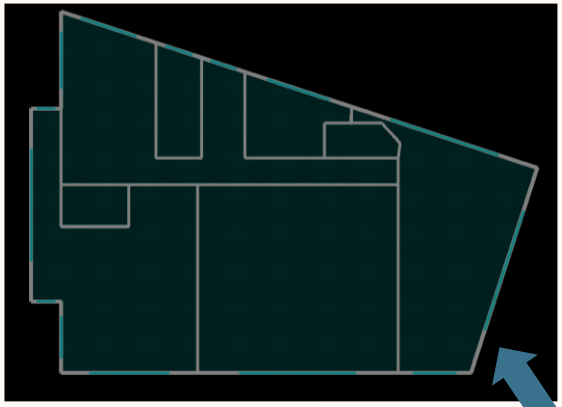
Developed eQUEST model of Foundation Building to estimate AHU energy use and ECMs savings.

AHU Zoning

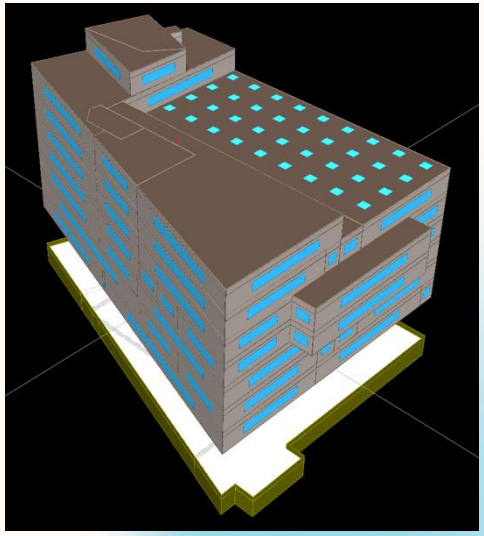
- zoning drawings found during the building walkthrough used to recreate floor-by-floor thermal zoning

Air Side Systems

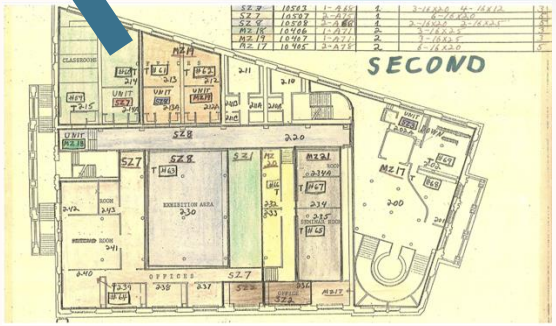
- Built HVAC systems in the model using available AHU information
- Modeled locked VAV AHUs as operating closer to constant volume systems.
- For units locked at 75% speed, fan power was adjusted using the affinity law:
  Fan Power = $(0.75)^3 = 0.42$, or about 42% of full-speed horsepower.



E-quest 2nd Floor Zoning



E-quest Building Model

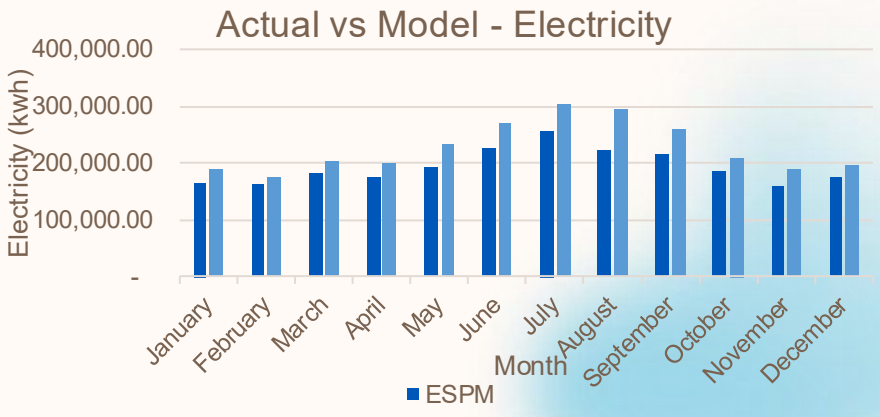


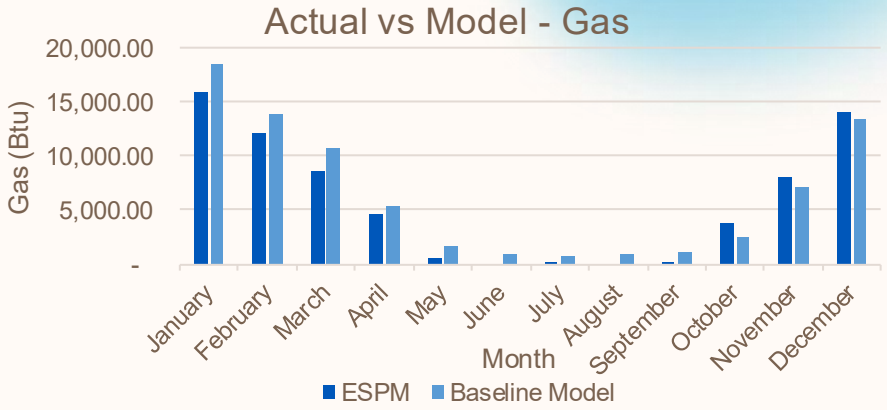

2nd Floor Zoning Drawing

# Foundation Building eQuest Model

- Adjusted occupancy levels, schedules assumptions to better match actual building operation.
- Model iteratively calibrating by comparing eQUEST outputs to expected AHU horsepower and adjusting static pressure inputs where needed.

| Source | Percent Differnce |
|---|---|
| Gas | 13.45 % |
| Electricity | 17.53 % |



# Implementations: ECM-1A, ECM-1B

| | | |
|---|---|---|
| ECM-1A | HVAC Scheduling:<br>Turn off AHUs 12 AM to 5 AM | Stand-alone time clock at each of 17 non-Siemens AHUs, interrupts fan starter at MCC. |
| ECM-1B | Night Space Temp Setbacks:<br>Set back space temp setpoints 10 PM to 7 AM (60 / 85 F) | Full DDC retrofit at 17 non-Siemens AHUs: controllers, sensors, valves, damper actuators, BMS network. |
| ECM 1A + 1B | HVAC Scheduling:<br>Turn off AHUs 12 AM to 5 AM<br>+ Setback space temp setpoints 10 PM to 12 AM and 5 AM to 7 AM (60/85) | Full DDC retrofit at 17 non-Siemens AHUs: controllers, sensors, valves, damper actuators, BMS network. |

**Just** ECM-1A HVAC Scheduling:

- **Stand-Alone Time Clock Installation:** The time clock is wired to interrupt the fan starter at the motor control center (MCC). Acts as a local control device that automatically turns off AHU during unoccupied hours.



OR

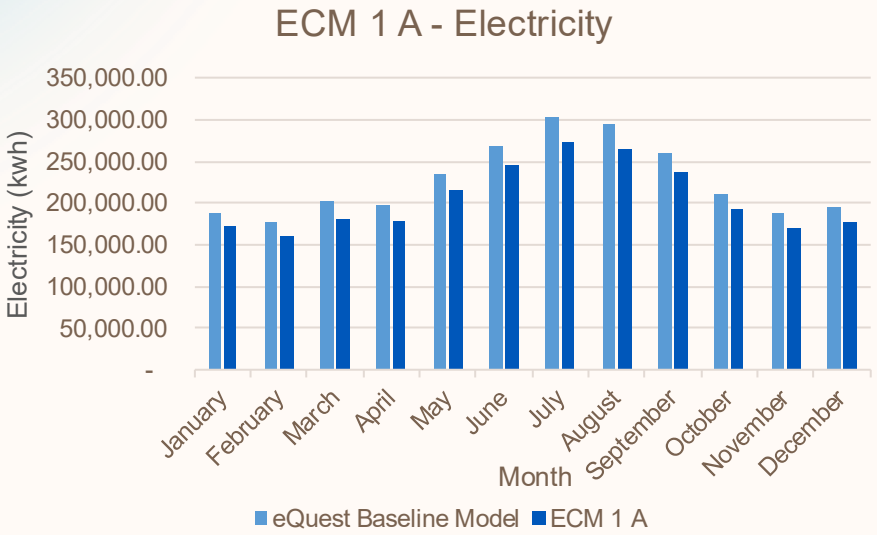
**Both** ECM-1B and ECM-1A:

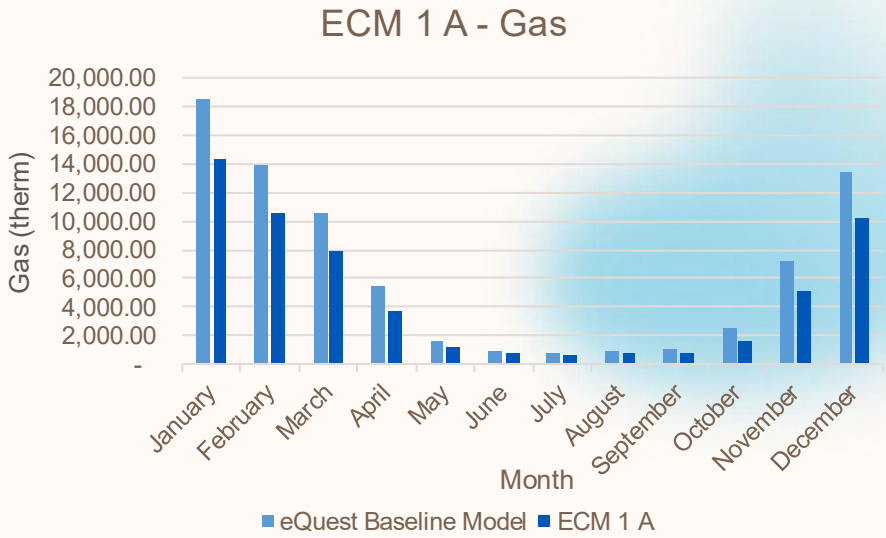
- **Full Direct Digital Control retrofit:** Includes controllers, sensors, valves, damper actuators, and BMS network connection.



# ECM-1A | HVAC Scheduling

Turn off AHUs 12 AM to 5 AM

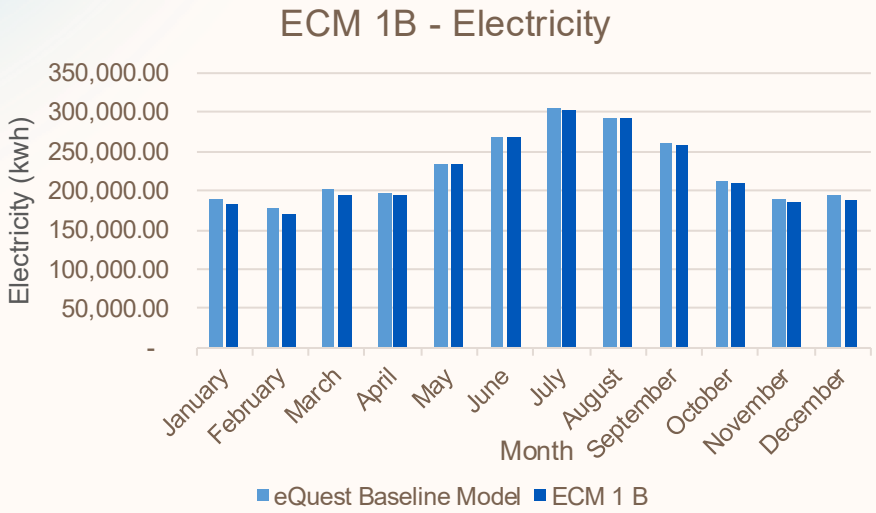


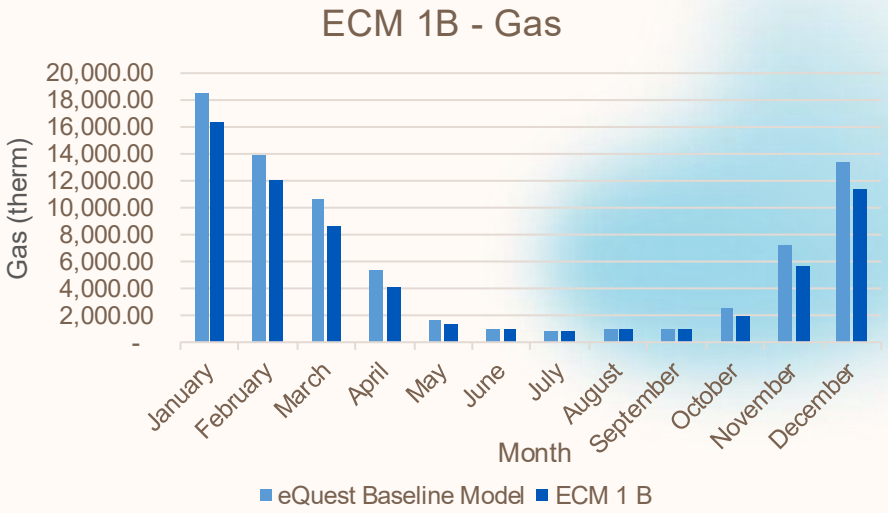




| Source | Reduction | Percent Differnce | Annual Cost Saving ($) | Annual Carbon Savings (mtCO2) |
|---|---|---|---|---|
| Electricity | 253,200.00 | 9.3% | 49,120.80 | 73.17 |
| Gas | 19,400.00 | 25.3% | 23,008.40 | 103.03 |
| Total | | | 72,129.20 | 176.20 |

# ECM-1B | HVAC Night Space Temperature Setbacks

60 °F Setback from 70 °F during heating & 85 °F Setback from 75 °F during cooling

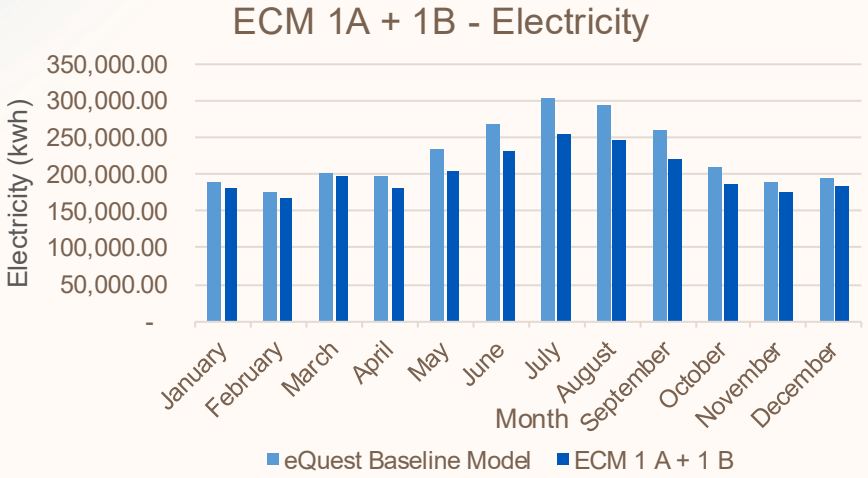


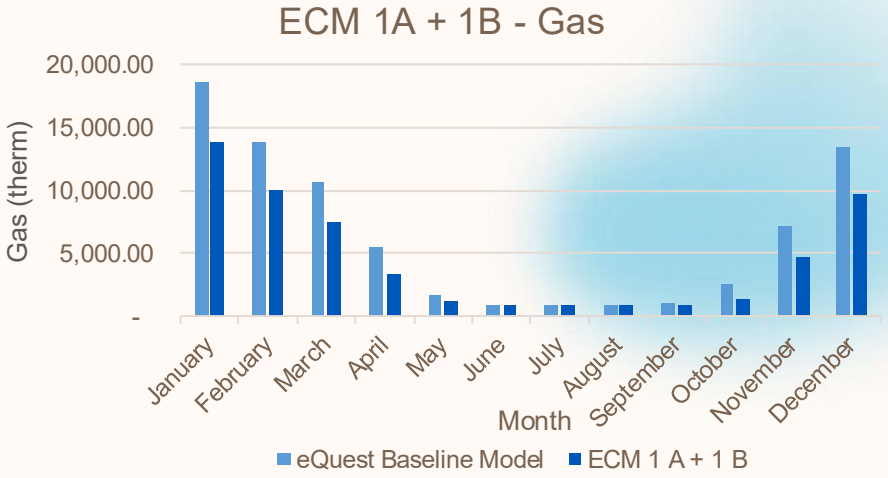




| Source | Reduction | Percent Differnce | Annual Cost Saving ($) | Annual Carbon Savings (mtCO2) |
|---|---|---|---|---|
| Electricity | 44,200.00 | 1.6% | 8,574.80 | 12.77 |
| Gas | 11,700.00 | 15.3% | 13,876.20 | 62.14 |
| Total | | | 22,451.00 | 74.91 |

# ECM-1A + 1 B | HVAC Night Space Temperature Setbacks

Turn off AHUs 12 AM to 5 AM and temperature setback setpoints from 10 PM to 12 AM and 5 AM to 7 AM




| Source | Reduction | Percent Differnce | Annual Cost Saving ($) | Annual Carbon Savings (mtCO2) |
|---|---|---|---|---|
| Electricity | 290,300.00 | 10.7% | 56,318.20 | 83.90 |
| Gas | 22,000.00 | 28.7% | 26,092.00 | 116.84 |
| Total | | | 82,410.20 | 200.74 |

# Implementation: ECM-2

| | |
|---|---|
| VAV Operation:<br>Operate VFD AHUs (AH3D, AH3B) as variable volume (currently constant volume) | Investigate and recommission 9 Carrier VFD VAV units to restore static pressure reset. |



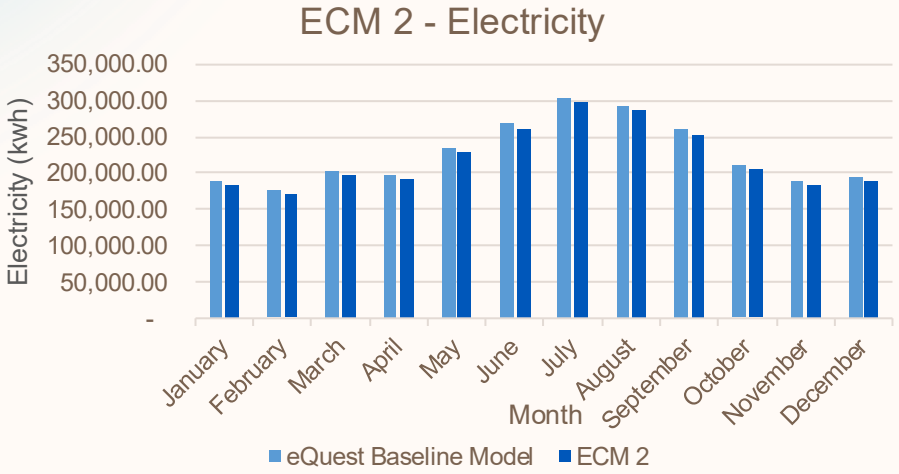
Controls Measures

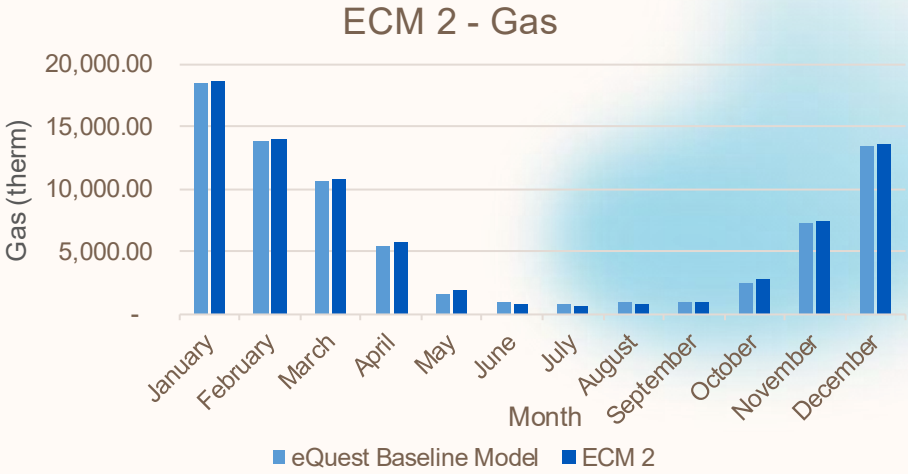
1. **Restore Static Pressure Reset**: Adjust controls to re-enable or correct static pressure reset so the fan speed can reduce when less airflow is needed.

2. **Investigate/Recommission VFD VAV units:** Investigate why the 9 Carrier VFD VAV units are not operating as intended. Check sensors, control programming, VFD settings, damper operation, and BMS communication. Recommission the units to confirm they are responding properly.

# ECM-2 | HVAC Night Space Temperature Setbacks

Operate existing Variable Frequency Drive Units as variable volume.

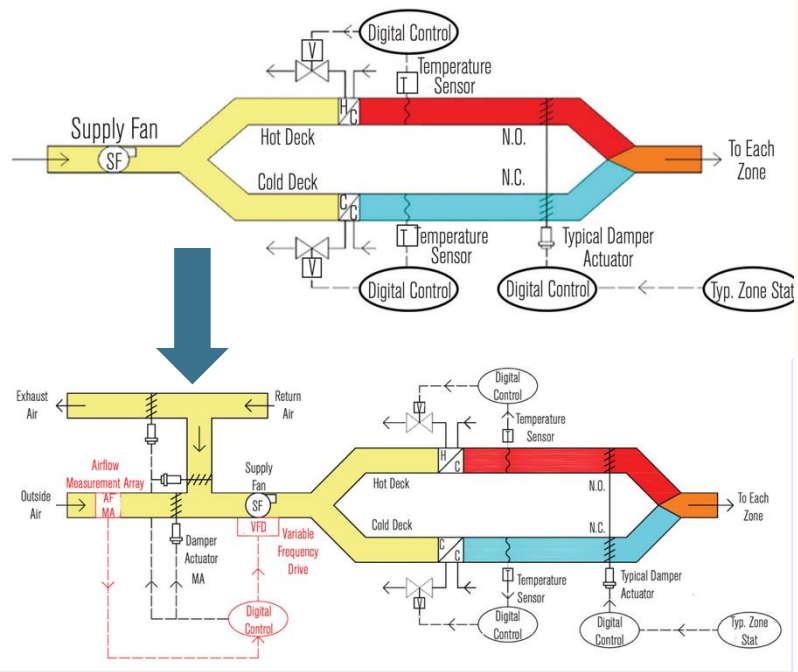



| Source | Reduction | Percent Differnce | Annual Cost Saving ($) | Annual Carbon Savings (mtCO2) |
|---|---|---|---|---|
| Electricity | 70,300.00 | 2.6% | 13,638.20 | 20.32 |
| Gas | (1,400.00) | -1.8% | (1,660.40) | (7.44) |
| Total | | | 11,977.80 | 12.88 |

# Implementation: ECM-3

| | |
|---|---|
| Convert MZ to VAV:<br>Convert multizone AHUs (MZ-1, 2, 3, 4, 5, 6, 7, 8, 18, 19, 20, 21) to variable volume | MZ-to-VAV retrofit on 12 multizone AHUs: VFDs, AFMAs, full DDC, zone damper separation. |




E. T. Westervelt, B. Clark, and M. Owen, "Exploring a multizone to variable volume HVAC controls retrofit," ASHRAE Journal, pp. 44–53, Sep. 2022. Available: WBDG, https://www.wbdg.org/FFC/ARMYCOE/COEDG/11-MZ-VV_Controls_ASHRAE_Journal_Sep_2022.pdf

Conventional multizone's operate with 2 decks: one hot, one cold. At the end they mix.

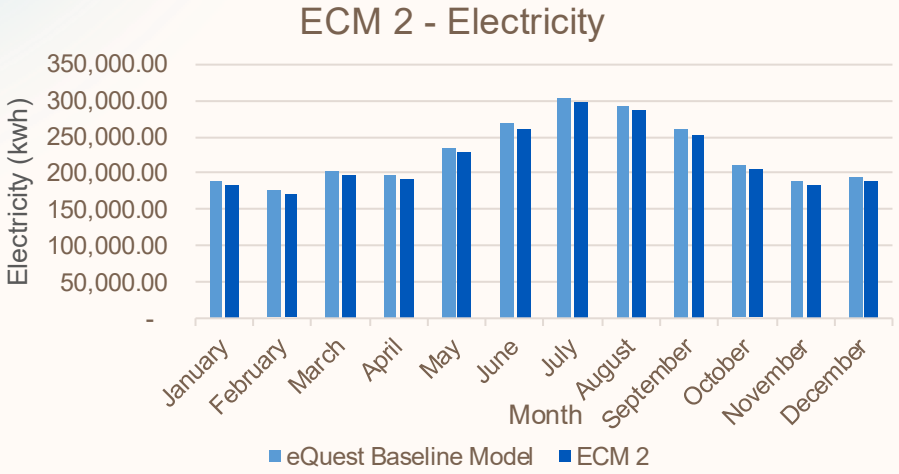
- Both decks typically operated continuously at a single speed, even when heating/cooling not needed.

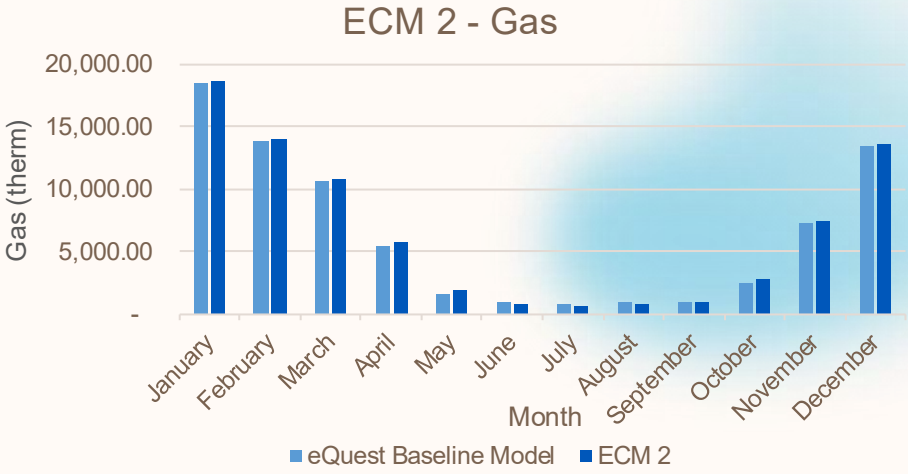
The Retrofit:

1. **Variable Frequency Drive (VFD) Installation:** Will allow for fan operation at reduced speed, lowering fan energy.
2. **Airflow Measurement Array (AFMA) Installation:** of an AFMA allows monitoring of outdoor air intake to supply the required amounts of ventilation and/or make-up air.
3. **Controls Programming:** Programming changes to implement the new functionality: variable fan speed control, outside airflow control, on/off heating/cooling coil control,

# ECM-3 | HVAC Night Space Temperature Setbacks

Convert multizone AHUs to variable volume




| Source | Reduction | Percent Differnce | Annual Cost Saving ($) | Annual Carbon Savings (mtCO2) |
|---|---|---|---|---|
| Electricity | 216,900.00 | 8.0% | 42,078.60 | 62.68 |
| Gas | 4,000.00 | 5.2% | 4,744.00 | 21.24 |
| Total | | | 46,822.60 | 83.93 |

# Savings and Payback

| Identification | | | | Baseline (Annual) | | | | Proposed (Annual) | | | | Savings (Annual) | | | | Capital and Payback (ROM) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Building | ECM | ECM Description | Model Includes | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Capital Expense (ROM) | After Rebate Expense (ROM) | Simple Payback |
| | | | | kWh | Therms | $ | MTCO2e | kWh | Therms | $ | MTCO2e | kWh | Therms | $ | MTCO2e | $ | $ | Years |
| FB | ECM-1A | HVAC Scheduling:<br>Turn off AHUs 12 AM to 5 AM | Whole Building | 2,720,100 | 76,700 | $618,666 | 1,193.4 | 2,466,900 | 57,300 | $546,536 | 1,017.2 | 253,200 | 19,400 | $72,129 | 176.2 | $60,000 | $60,000 | 0.8 yrs |
| FB | ECM-1B | Night Space Temp Setbacks:<br>Set back space temp setpoints 10 PM to 7 AM (60 / 85 F) | Whole Building | | | | | 2,675,900 | 65,000 | $596,215 | 1,118.4 | 44,200 | 11,700 | $22,451 | 74.9 | $600,000 | $600,000 | 26.7 yrs |
| FB | ECM 1A + 1B | HVAC Scheduling:<br>Turn off AHUs 12 AM to 5 AM<br>+ Setback space temp setpoints 10 PM to 12 AM and 5 AM to 7 AM (60/85) | Whole Building | | | | | 2,429,800 | 54,700 | $536,255 | 992.6 | 290,300 | 22,000 | $82,410 | 200.7 | $600,000 | $600,000 | 7.3 yrs |
| FB | ECM-2 | VAV Operation:<br>Operate VFD AHUs (AHU-X, X, X, X) as variable volume (currently constant volume) | Whole Building | | | | | 2,649,800 | 78,100 | $606,688 | 1,180.5 | 70,300 | (1,400) | $11,978 | 12.9 | - | - | - |
| FB | ECM-3 | Convert MZ to VAV:<br>Convert multizone AHUs (AHU-X, X, X, X, X) to variable volume | Whole Building | | | | | 2,503,200 | 72,700 | $571,843 | 1,109.4 | 216,900 | 4,000 | $46,823 | 83.9 | $1,000,000 | $1,000,000 | 21.4 yrs |

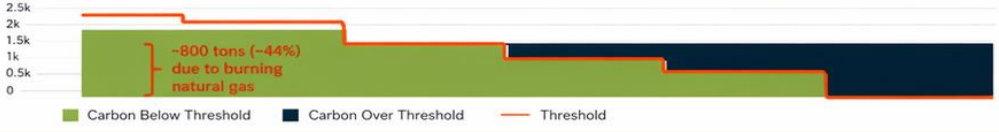
# Avoiding the Penalties

2025 Total Emissions

1,028 $tCO_2e$

**Annual Carbon Threshold Summary**

| $tCO_2e$/yr | 2024–2029 | 2030–2034 | 2035–2039 | 2040–2049 | 2050+ |
|---|---|---|---|---|---|
| Emissions | 1,839 | 1,322 | 1,322 | 1,322 | 1,322 |
| **Threshold** | 2,335 | 1,387 | 995 | 577 | 0 |
| **Est. Penalty** ($/yr) | $0 | $0 | $87,661 | $199,674 | $354,310 |




$87,661 / yr

LL97 penalties expected by 2035

Implement

ECM 1A
or
ECM 1A + 1B

to avoid penalties until 2040

# Next Steps and Recommendations

### Pursue more aggressive decarbonization strategies

- The 2040 target of 34 $tCO_2e$ is difficult to reach through AHU optimization alone.
- Even after implementing the proposed AHU-focused ECMs, the building is still projected to exceed future emissions limits. We need to **Expand beyond AHU optimization**

### Restore HVAC data streams and develop a data history

- **Restore data collection** from HVAC systems, including fan operation, VFD speed, zone temperatures, static pressure, valve positions, schedules, and energy use.
- Maintaining historical BMS data would make it easier to identify operational issues, verify ECM savings, track performance over time, and support future energy model calibration.

02

# 41 CS

# 41CS Lab Ventilation Audit (Fall 2025)



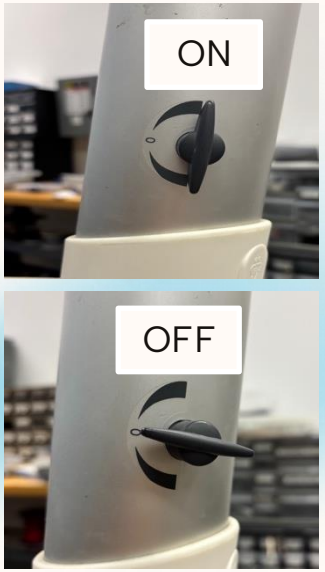
Inactive Open Fume Hood in Lab 601



Testing Ventilation in Lab LL219



Point Exhaust




Snorkel Hood On/Off Control

- Checked every lab in 41CS to check vents, fume hoods, and snorkels that are left on – even when not in use
- Summed energy savings from turning off inactive ventilation equipment in labs
- **Savings: 13,000CFM**

# Zone Analysis: Lab Differential Pressure (AHU-4/5)

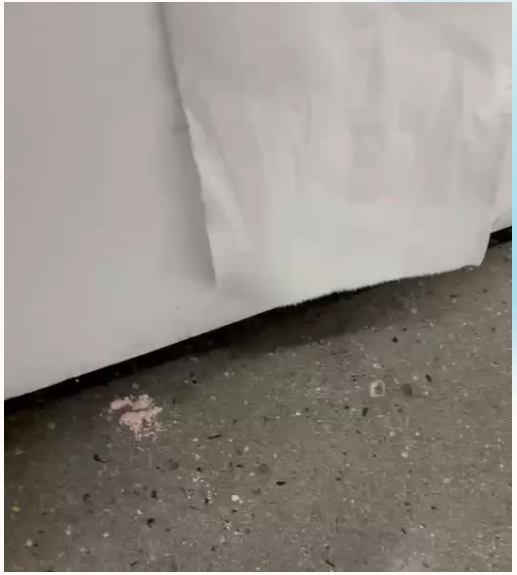
**12 Labs are Positively Pressured (based on BMS)**

| LAB | Differential Setpoint | Actual Differential |
|---|---|---|
| 402 | -100 | 220 |
| 404 N | -288 | 1068 |
| 404 S | -288 | 596 |
| 702 | -400 | 200 |
| 703 | -100 | 500 |
| 704 | -200 | 12 |
| 705 | -320 | 8 |
| 706 | -400 | 948 |
| LL218 | -300 | 1196 |
| LL219B | -400 | 0 |
| LL220 | -400 | 0 |
| LL221 | -400 | 0 |
| LL222A | -262 | 112 |
| LL224 | -452 | 584 |
| LL224A | -352 | 668 |

**Even labs that are Negatively Pressured (based on BMS) were tested and shown to be Positive**



701
BMS Setpoint: -300 CFM
BMS Actual: -200 CFM
POSITIVELY PRESSURIZED



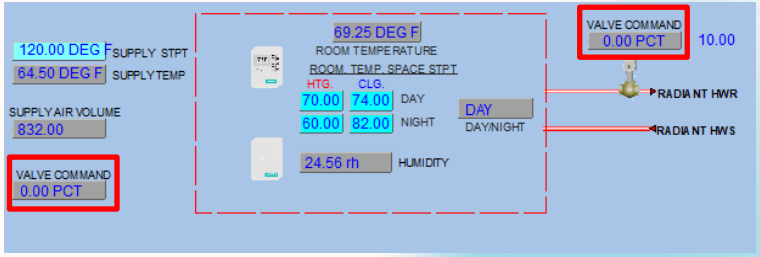
704
BMS Setpoint: -200 CFM
BMS Actual: -108 CFM
POSITIVELY PRESSURIZED

# Looking for Explanations in the TEC Manual revealed more issues…

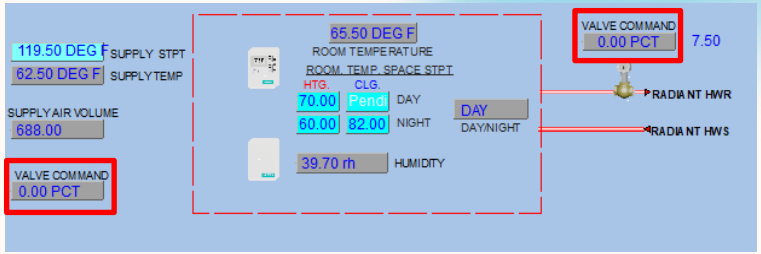
### TEC Operation vs Reality

| | What supposed to happen | Reality |
|---|---|---|
| **Pressurization** | Dampers modulate to meet pressure setpoints | Setpoints aren't met and labs are (supposedly) positively pressurized |
| **Temperature control** | Heating valves modulate to meet discharge setpoint | Heat valves don't modulate at all |

### Lab 702 Heat Valves




November 2025

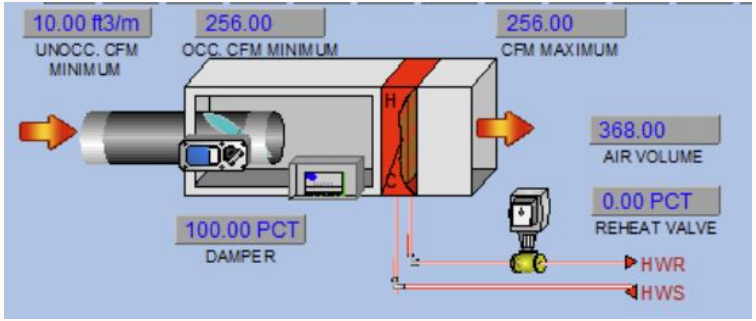



March 2026

7th Floor Labs with 0 PCT valves across 2017, 2019, 2025 BMS screenshots:
701, 702, 703, 704, 706, 708, 710

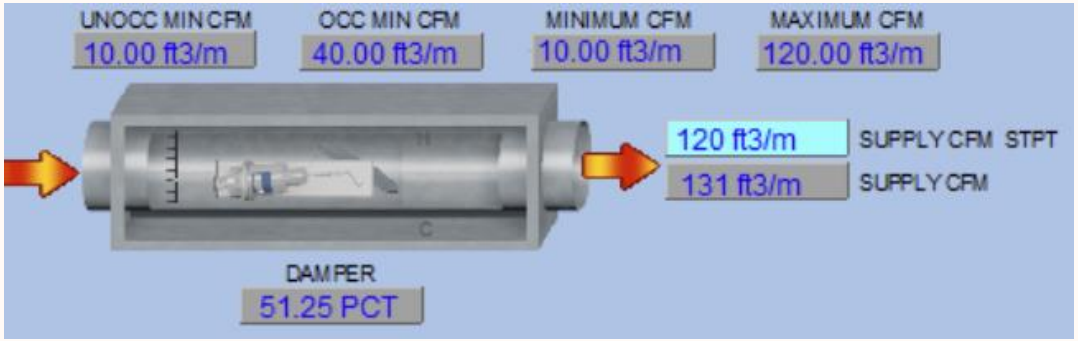
# Labs weren't the only issue...looked at classroom and offices (AHU-3/6)

AHU-3: Room 106 (Classroom) – CFM was greater than max and room was unoccupied




Occupancy: Unocc
Max CFM: 256
Supply CFM: 368 (+112 than setpoint)

AHU-6: Room 412 (Office) – Setpoint set at max CFM despite being unoccupied and CFM was greater than the setpoint



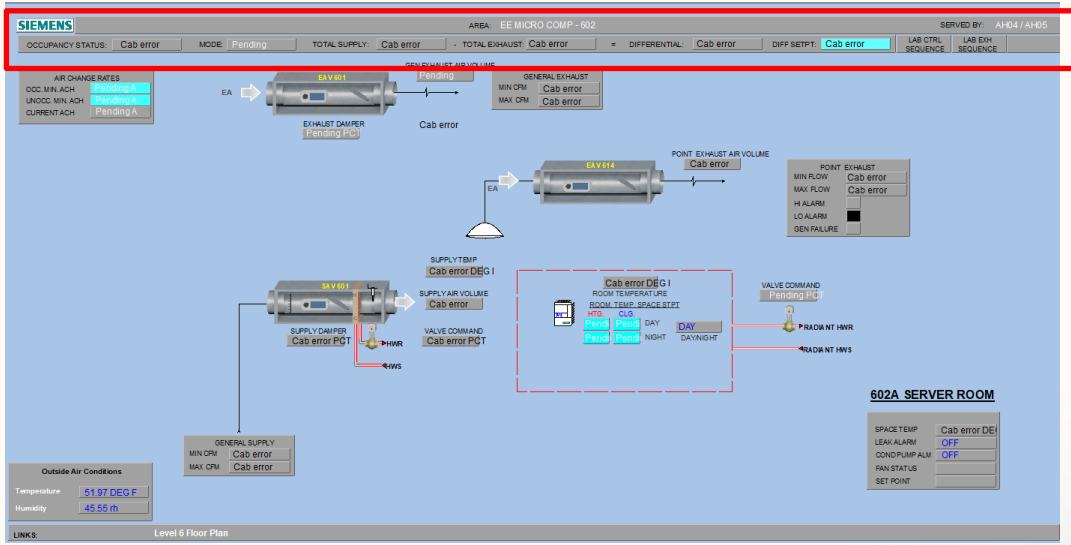

Occupancy: Unocc
Max CFM: 120
Supply Setpoint: 120
Supply CFM: 131 (+11 than setpoint)

# 8760Hr Model Inputs

Fall/Spring Semester (2025/26):

- Many sensors are broken – so there is no way of quantifying savings and if pressurization is negative/positive
  - Ex: All of Floor 6 is “Cab Error”

- Appended 2019 AHU Primer Document with 2026 values
  - 2026 CFM values are ~6,000-10,000CFM higher

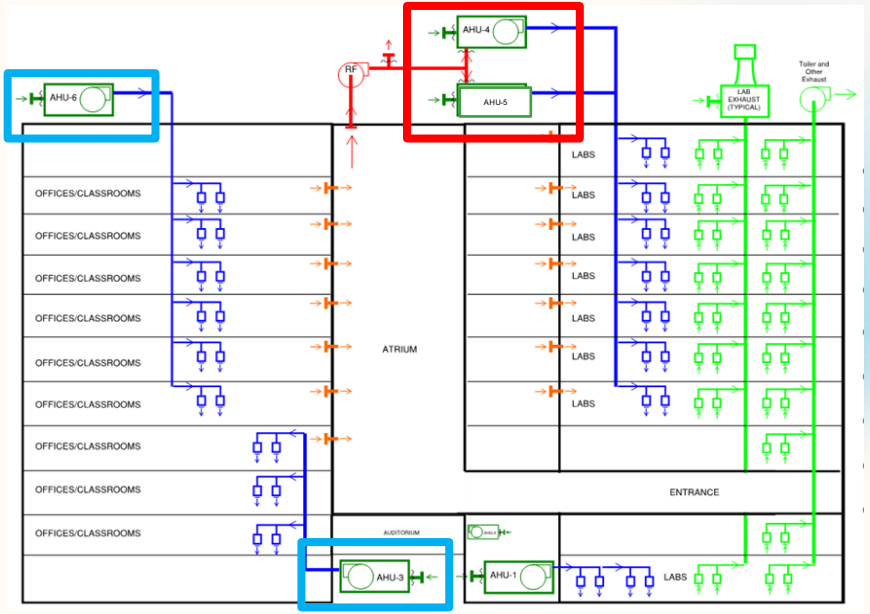



- **So… we checked design values and 2020 trends to see how much energy reduction is actually achievable.**
- **Context: in 2020, a lot of CAP work had been completed and those savings were maintained.**

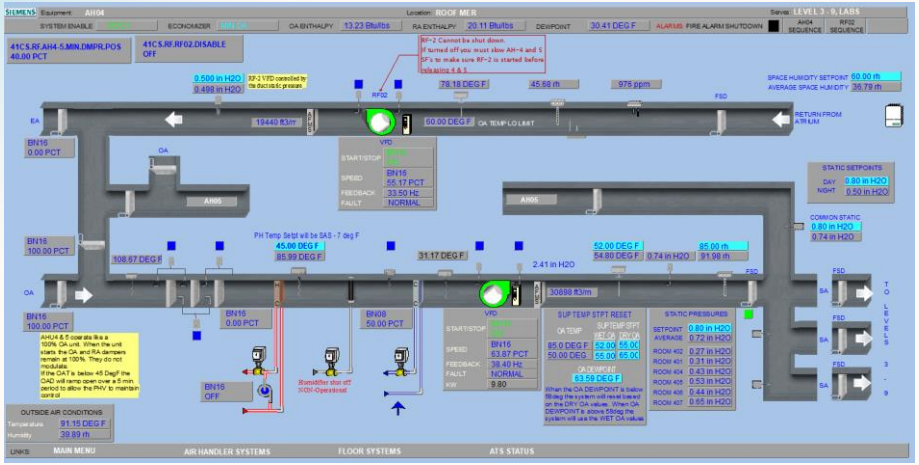
# Developing an Updated AHU Model

- **Created a theoretical model for Air Handling Units (AHU) 3,4,5,6**
  - 3,6 service **offices/classrooms**
  - 4,5 service the 41CS **labs**

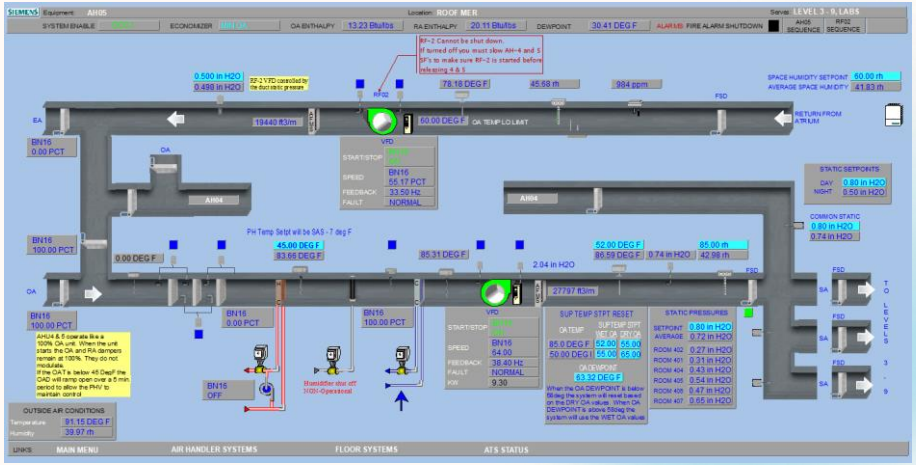
- Modeled 41CS with an 8760hr (1yr) Model
- Modeled energy and **cost savings of $242,000** if we return to 2020 settings (and beyond)



# Building Management System (BMS)



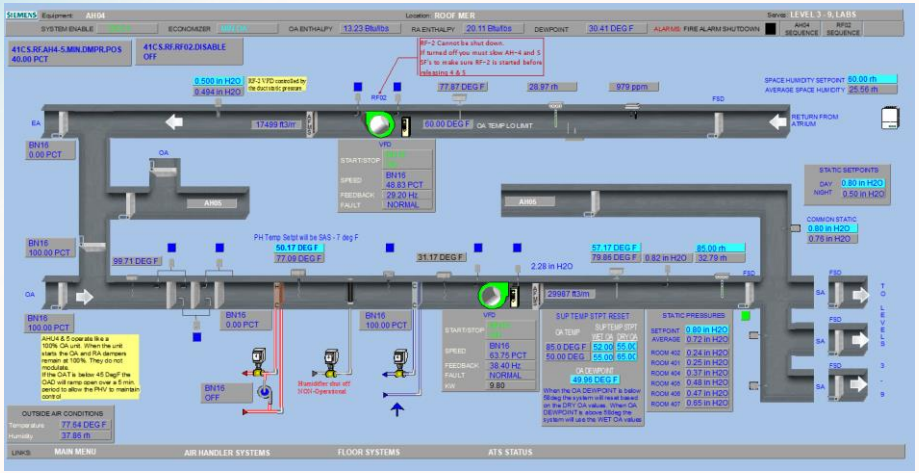
AHU4



AHU5

- AHU4 and AHU5 are connected
- They are **MIXED AIR units**
- Same Return Air
- Different Supply Air

# March 10, 2026 BMS Screenshots

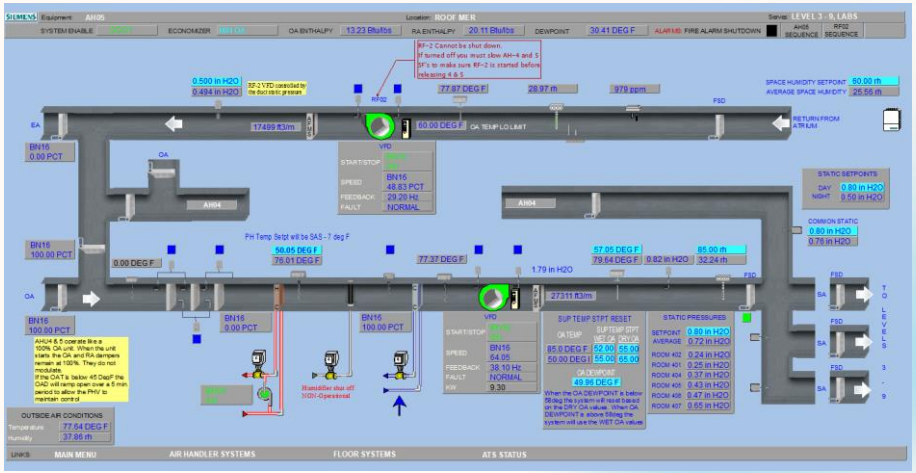


AHU4

Supply: 29,987CFM
**Return**: 17,499CFM (**29%**)
**OA**: 21,237.5CFM (**71%**)

*Per AHU: 8,749.5CFM*

**Return Air is split between AHU 4 and 5**



AHU5

Supply: 27,311CFM
**Return**: 17,499CFM (**32%**)
**OA**: 18,561.5CFM (**68%**)

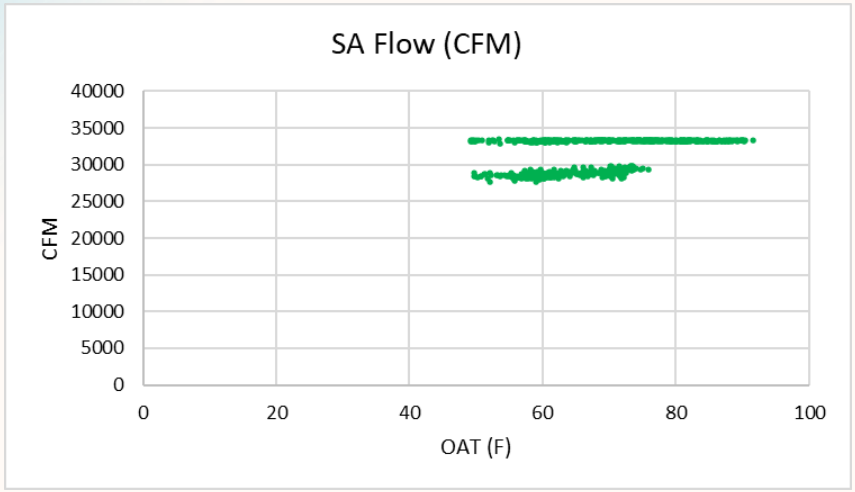
# Outside Air (OA) vs Return Air (RA) AHU4

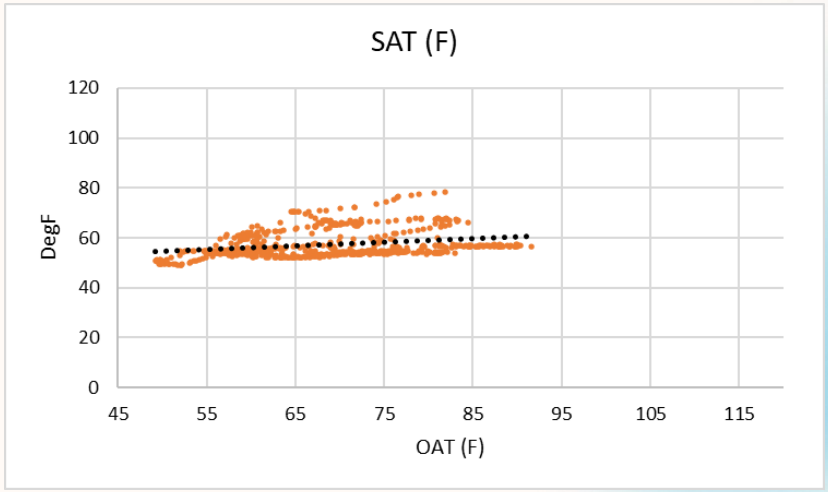
| DATE | AHU4 RA % | AHU5 RA % | AHU4 OA % | AHU5 OA % |
|---|---|---|---|---|
| February 21, 2026 | 29 | 32 | 71 | 68 |
| March 10, 2026 | 29 | 32 | 71 | 68 |
| April 1, 2026 (Nighttime) | 40 | 43 | 60 | 57 |
| April 15, 2026 | 31 | 35 | 69 | 65 |

- **Return Air: ~30%**
- **Outside Air: ~70%**

# 8760Hr (1 year) Model

2025

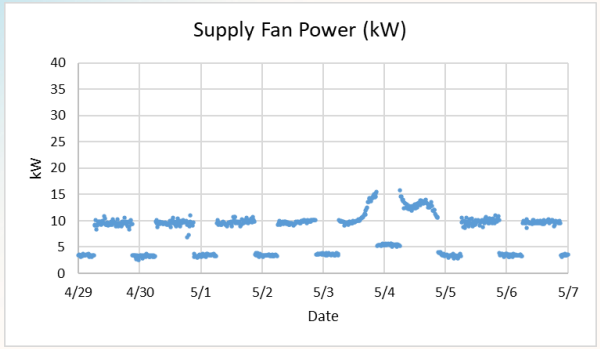


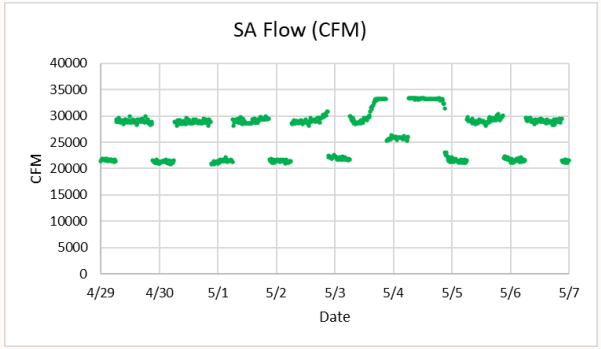



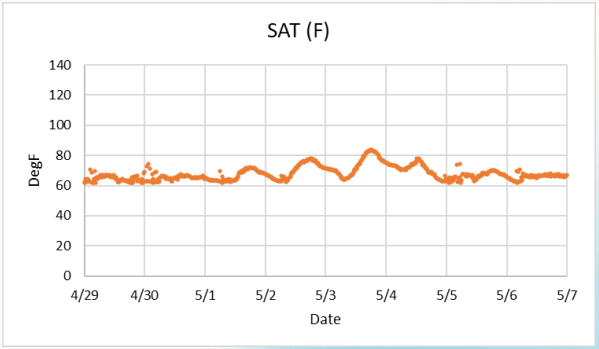

- Looked at current (2025-26) and past trends (2020)
- Updated an 8760Hr (1yr) model
- Baseline year: 2025

- **“Improved” energy costs by reducing CFM compared to 2025**

# Time Trends: AHU4 2020



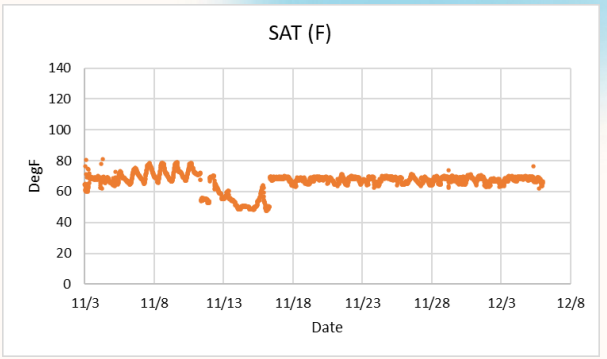




- 4/29/2020 – 5/7/2020

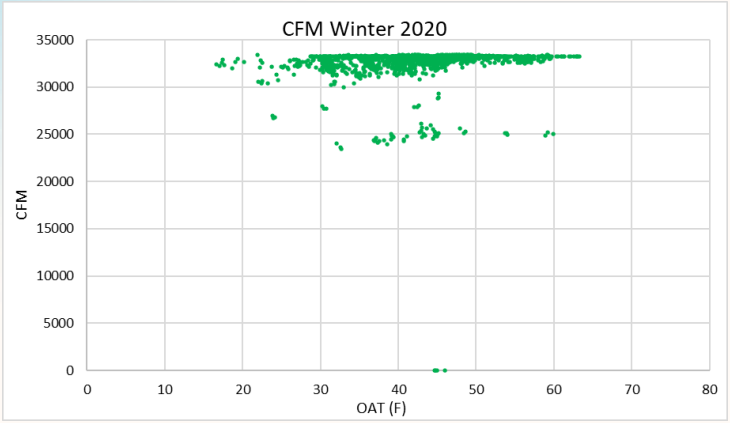
*Setbacks start 8-10PM and end 6AM*

- kW
  - Clear nighttime setbacks (10 and 4)
- CFM
  - **Clear nighttime setbacks (30,000 and 21,500)**
- SAT
  - No clear nighttime setbacks

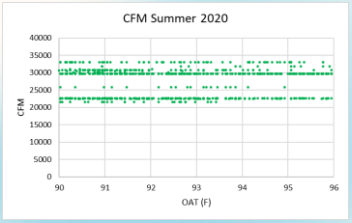


# 2020 OAT Data: AHU4 CFM

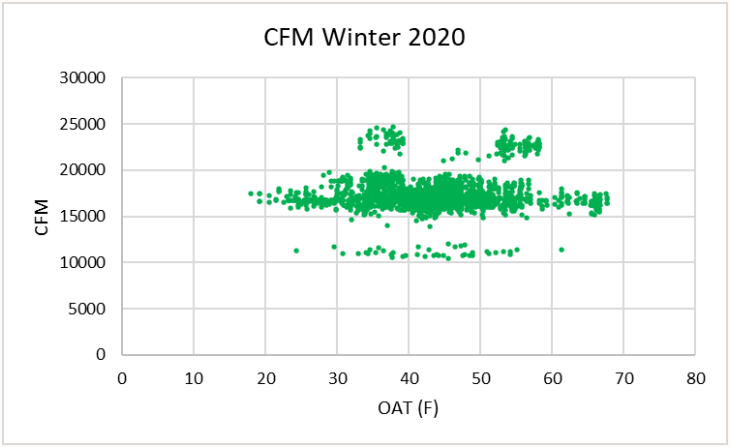
**Everything is Daytime**



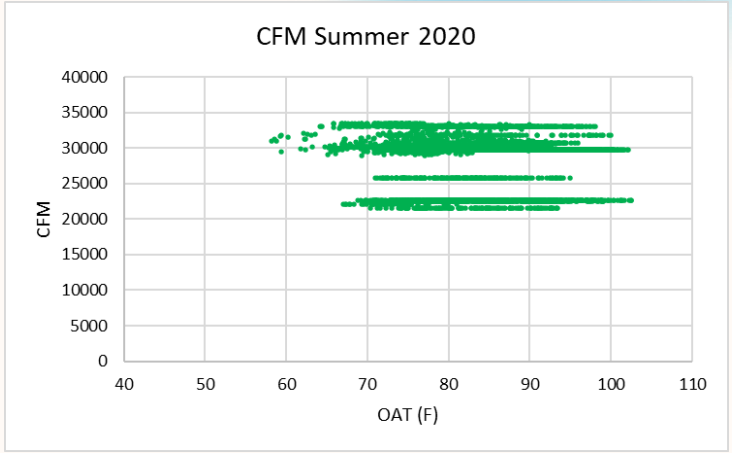

Feb 2020 data
0 deg: 34,000CFM
52 deg: 34,000CFM
57 deg:  34,000CFM



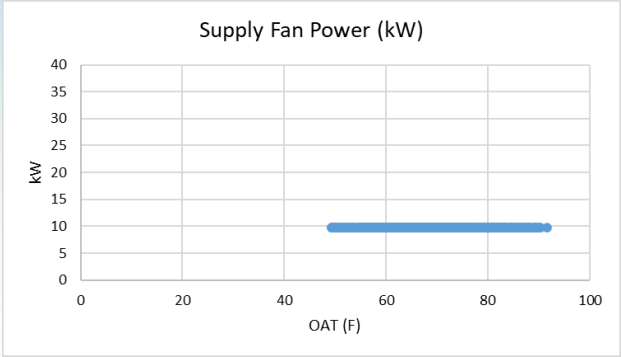

Jun-Aug 2020 data
95 deg: 30,000CFM

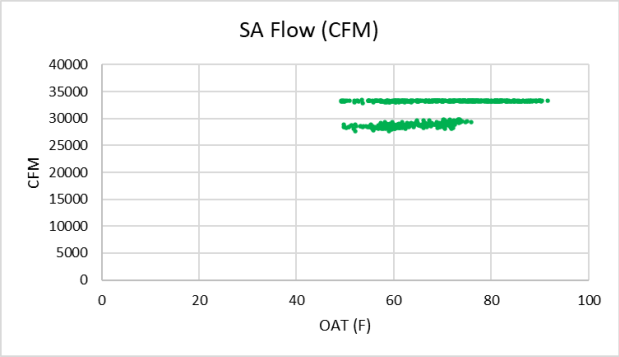


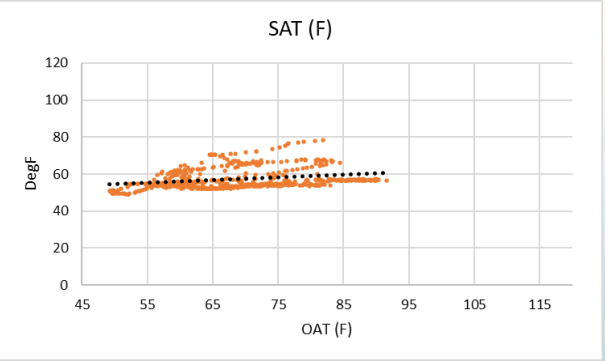

Dec2019-Jan2020
**0 deg: 17,000CFM**
**52 deg: 20,000CFM**
**57 deg: 20,000CFM**



# Outside Air Trends: AHU4 2025




- 4/27/2025 – 5/3/2025

*Setbacks start 8-10PM and end 6AM*

- kW
  - Sensor is broken on BMS
- CFM
  - Clear nighttime setbacks (34,000 and 29,000)
- SAT
  - Constant at ~58F

**+4,000 DAYTIME (as compared to 2020)**

**+8,000 NIGHTTIME (as compared to 2020)**

# Outside Air Trends: 2026

**TOTAL CFM**

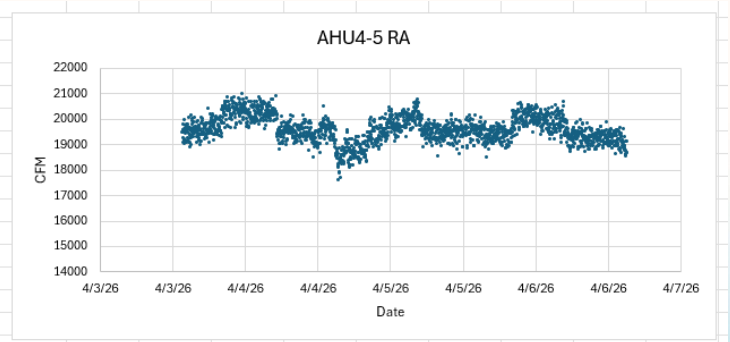
- Supply
  - ON: 58,500
  - OFF: 47,300
- Return
  - 19,650
- OA
  - ON: 38,850
  - OFF: 27,650

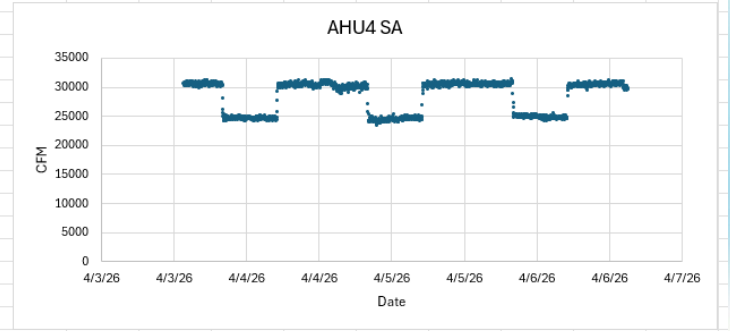
AHU4

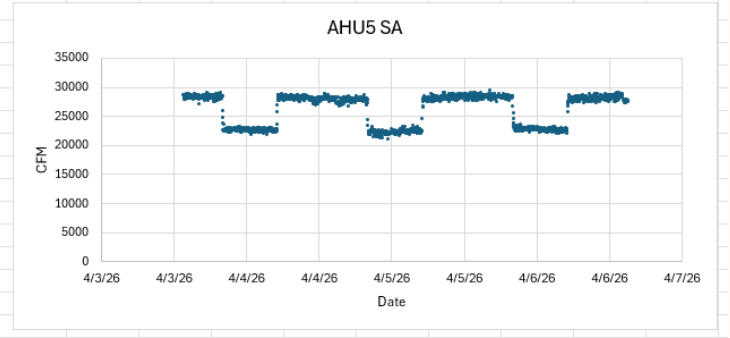
- ON: 30,500
- OFF: 25,000

**+500 DAYTIME**
**(as compared to 2020)**

**+4,000 NIGHTTIME**
**(as compared to 2020)**



# Current Operation vs Design Values

**Repeated trend analysis with other AHUs and compared CFM to design values**

| AHU | Design CFM | Current CFM | Difference |
|---|---|---|---|
| AHU-3 | 24,600 | 28,000 | **+3,400** |
| AHU-4 | 29,700 | 31,000 | **+1,300** |
| AHU-5 | 29,700 | 35,000 | **+5,300** |
| AHU-6 | 27,500 | 35,000 | **+7,500** |

| ITEM | LOCATION | AREA SERVED | CFM |
|---|---|---|---|
| AHU - 1 | SUB-CELLAR | SUB-CELLAR | 22,600 |
| AHU - 2 | SUB-CELLAR | CELLAR | 4,200 |
| AHU - 3 | SUB-CELLAR | CELLAR 1-3 FLOOR | 24,600 |
| AHU - 4 | ROOF | 3-9 FLOOR | 29,700 |
| AHU - 5 | ROOF | 3-9 FLOOR | 29,700 |
| AHU - 6 | ROOF | 3-8 FLOOR | 27,500 |
| HV - 1 | SUB-CELLAR | MER | 3,000 |

**AHU CFM Design Values**

# 41CS Energy Cost Model (ECM1)

**ECM1: Nighttime Setbacks (Reduce 22:00-6:00)**



- **1A: Complete** nighttime shut off (0 CFM at night)
  - AHU3/6
- **1B: Aggressive** nighttime reduction
  - AHU3/6 (50% reduction)
  - AHU4/5 reduces by 13,000 CFM (24% reduction)
- **1C: Moderate** nighttime reduction
  - AHU3/6 (20% reduction)
  - AHU4/5 reduces by 7,500 CFM (12% reduction)

# ECM1A: Energy and Cost Savings

Energy Rates:
Electric: $0.19/kWh
Natural Gas: $11.86/mmbtu

| | Mechanical Cooling Load (Tons) | Fan Motor (kWh) | Cooling Elec Energy (CHW or DX) (kWh) | Fuel Input to Heating Plant (kbtu) | Fuel Cost – Total($) | Electrical Energy – Total (kW) | Max Monthly Elec (KW) | Electricity Cost – Total ($) | **Total Elec and Fuel Cost ($)** |
|---|---|---|---|---|---|---|---|---|---|
| **AHU3** | 0 | 77,000 | 57,000 | 1,300,000 | 15,000 | 133,000 | 0 | 26,000 | **41,000** |
| **AHU6** | 2 | 92,000 | 45,000 | 1,170,000 | 14,000 | 137,000 | 0 | 26,000 | **40,000** |

**Complete Nighttime Shutoff**
**200 annual MTCO2e Saved**
**$81k Cost Savings**

## ECM1B BASELINE AHU4/5

| Part Load Conditions: AHU SAT Reset | | |
|---|---|---|
| Supply Air Temp (°F) | AHU SAT (F) | Average AHU SA CFM |
| 95 | 52.0 | 64,253 |
| 57 | 52.0 | 63,869 |
| 52 | 63.0 | 63,486 |
| 0 | 73.0 | 63,102 |

2025 CFM

| Unit Schedule | | |
|---|---|---|
| Day of Week | Start Hour | Stop Hour |
| Monday | 0 | 24 |
| Tuesday | 0 | 24 |
| Wednesday | 0 | 24 |
| Thursday | 0 | 24 |
| Friday | 0 | 24 |
| Saturday | 0 | 24 |
| Sunday | 0 | 24 |

| Space Heating Setpoint Setback Factor | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 22 | 6 |
| Weekend | 22 | 6 |
| Space Heating Setpoint Setback | 0.0 | °F |

| Morning Start-Up | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 6 | 22 |
| Weekend | 6 | 22 |
| CFM Multiplier | 1.00 | |

No Setbacks

## ECM1B PROPOSED

| Part Load Conditions: AHU SAT Reset | | |
|---|---|---|
| Supply Air Temp (°F) | AHU SAT (F) | Average AHU SA CFM |
| 95 | 52 | 64,253 |
| 57 | 52 | 63,869 |
| 52 | 63 | 63,486 |
| 0 | 73 | 63,102 |

Same Daytime CFM

| Unit Schedule | | |
|---|---|---|
| Day of Week | Start Hour | Stop Hour |
| Monday | 0 | 24 |
| Tuesday | 0 | 24 |
| Wednesday | 0 | 24 |
| Thursday | 0 | 24 |
| Friday | 0 | 24 |
| Saturday | 0 | 24 |
| Sunday | 0 | 24 |

| Space Heating Setpoint Setback Factor | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 22 | 6 |
| Weekend | 22 | 6 |
| Space Heating Setpoint Setback | 0 | °F |

No Setback

| Morning Start-Up | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 4 | 22 |
| Weekend | 4 | 22 |
| CFM Multiplier | 0.80 | °F |

13,000 CFM Night Setback

# ECM1B: Energy and Cost Savings

Energy Rates:
Electric: $0.19/kWh
Natural Gas: $11.86/mmbtu

| | Mechanical Cooling Load (Tons) | Fan Motor (kWh) | Cooling Elec Energy (CHW or DX) (kWh) | Fuel Input to Heating Plant (kbtu) | Fuel Cost – Total($) | Electrical Energy – Total (kW) | Max Monthly Elec (KW) | Electricity Cost – Total ($) | **Total Elec and Fuel Cost ($)** |
|---|---|---|---|---|---|---|---|---|---|
| **AHU3** | 0 | 54,000 | 18,000 | 347,000 | 4,000 | 72,000 | 0 | 14,000 | **18,000** |
| **AHU6** | 0 | 57,000 | 13,000 | 344,000 | 4,000 | 69,000 | 0 | 13,000 | **18,000** |
| **AHU4/5** | 2 | 50,000 | 17,000 | 570,000 | 7,000 | 69,000 | 0 | 13,000 | **20,000** |

**Aggressive Nighttime Shutoff**
**100 annual MTCO2e Saved**
**$56k Cost Savings**

## ECM1C BASELINE AHU4/5

| Part Load Conditions: AHU SAT Reset | | |
|---|---|---|
| Supply Air Temp (°F) | AHU SAT (F) | Average AHU SA CFM |
| 95 | 52.0 | 64,253 |
| 57 | 52.0 | 63,869 |
| 52 | 63.0 | 63,486 |
| 0 | 73.0 | 63,102 |

2025 CFM

| Unit Schedule | | |
|---|---|---|
| Day of Week | Start Hour | Stop Hour |
| Monday | 0 | 24 |
| Tuesday | 0 | 24 |
| Wednesday | 0 | 24 |
| Thursday | 0 | 24 |
| Friday | 0 | 24 |
| Saturday | 0 | 24 |
| Sunday | 0 | 24 |

| Space Heating Setpoint Setback Factor | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 22 | 6 |
| Weekend | 22 | 6 |
| Space Heating Setpoint Setback | 0.0 | °F |

| Morning Start-Up | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 6 | 22 |
| Weekend | 6 | 22 |
| CFM Multiplier | 1.00 | |

No Setbacks

## ECM1C PROPOSED

| Part Load Conditions: AHU SAT Reset | | |
|---|---|---|
| Supply Air Temp (°F) | AHU SAT (F) | Average AHU SA CFM |
| 95 | 52 | 64,253 |
| 57 | 52 | 63,869 |
| 52 | 63 | 63,486 |
| 0 | 73 | 63,102 |

Same Daytime CFM

| Unit Schedule | | |
|---|---|---|
| Day of Week | Start Hour | Stop Hour |
| Monday | 0 | 24 |
| Tuesday | 0 | 24 |
| Wednesday | 0 | 24 |
| Thursday | 0 | 24 |
| Friday | 0 | 24 |
| Saturday | 0 | 24 |
| Sunday | 0 | 24 |

| Space Heating Setpoint Setback Factor | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 22 | 6 |
| Weekend | 22 | 6 |
| Space Heating Setpoint Setback | 0 | °F |

No Setback

| Morning Start-Up | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 4 | 22 |
| Weekend | 4 | 22 |
| CFM Multiplier | 0.88 | °F |

7,500 CFM Night Setback

# ECM1C: Energy and Cost Savings

Energy Rates:
Electric: $0.19/kWh
Natural Gas: $11.86/mmbtu

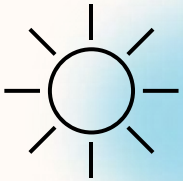
| | Mechanical Cooling Load (Tons) | Fan Motor (kWh) | Cooling Elec Energy (CHW or DX) (kWh) | Fuel Input to Heating Plant (kbtu) | Fuel Cost – Total($) | Electrical Energy – Total (kW) | Max Monthly Elec (KW) | Electricity Cost – Total ($) | **Total Elec and Fuel Cost ($)** |
|---|---|---|---|---|---|---|---|---|---|
| **AHU3** | 0 | 27,000 | 7,000 | 87,000 | 1,000 | 34,000 | 0 | 6,600 | **7,600** |
| **AHU6** | 0 | 20,000 | 3,000 | 43,000 | 500 | 23,000 | 0 | 4,400 | **5,000** |
| **AHU4/5** | 2 | 31,000 | 10,000 | 333,000 | 4,000 | 42,000 | 0 | 8,000 | **12,000** |

**Moderate Nighttime Shutoff**
**100 annual MTCO2e Saved**
**$25k Cost Savings**

# 41CS Energy Cost Model (ECM2)

**ECM 2: Zone Level Optimization (Day and Night)**

- AHU3/6:  3000-5000 CFM reduction
- AHU4/5:  13,000 CFM reduction



## ECM2 BASELINE AHU4/5

| Part Load Conditions: AHU SAT Reset | | |
|---|---|---|
| Supply Air Temp (°F) | AHU SAT (F) | Average AHU SA CFM |
| 95 | 52.0 | 64,253 |
| 57 | 52.0 | 63,869 |
| 52 | 63.0 | 63,486 |
| 0 | 73.0 | 63,102 |

2025 CFM

| Unit Schedule | | |
|---|---|---|
| Day of Week | Start Hour | Stop Hour |
| Monday | 0 | 24 |
| Tuesday | 0 | 24 |
| Wednesday | 0 | 24 |
| Thursday | 0 | 24 |
| Friday | 0 | 24 |
| Saturday | 0 | 24 |
| Sunday | 0 | 24 |

| Space Heating Setpoint Setback Factor | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 22 | 6 |
| Weekend | 22 | 6 |
| Space Heating Setpoint Setback | 0.0 | °F |

| Morning Start-Up | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 6 | 22 |
| Weekend | 6 | 22 |
| CFM Multiplier | 1.00 | |

No Setbacks

## ECM2 PROPOSED

| Part Load Conditions: AHU SAT Reset | | |
|---|---|---|
| Supply Air Temp (°F) | AHU SAT (F) | Average AHU SA CFM |
| 95 | 52 | 51,253 |
| 57 | 52 | 50,869 |
| 52 | 63 | 50,486 |
| 0 | 73 | 50,102 |

-13,000CFM Daytime

| Unit Schedule | | |
|---|---|---|
| Day of Week | Start Hour | Stop Hour |
| Monday | 0 | 24 |
| Tuesday | 0 | 24 |
| Wednesday | 0 | 24 |
| Thursday | 0 | 24 |
| Friday | 0 | 24 |
| Saturday | 0 | 24 |
| Sunday | 0 | 24 |

| Space Heating Setpoint Setback Factor | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 22 | 6 |
| Weekend | 22 | 6 |
| Space Heating Setpoint Setback | 0 | °F |

No Setback

| Morning Start-Up | | |
|---|---|---|
| | Start Hour | Stop Hour |
| Weekday | 4 | 22 |
| Weekend | 4 | 22 |
| CFM Multiplier | 1.00 | |

No Setback (-13,000CFM total)

# ECM2: Energy and Cost Savings

Energy Rates:
Electric: $0.19/kWh
Natural Gas: $11.86/mmbtu

| | Mechanical Cooling Load (Tons) | Fan Motor (kWh) | Cooling Elec Energy (CHW or DX) (kWh) | Fuel Input to Heating Plant (kbtu) | Fuel Cost – Total($) | Electrical Energy – Total (kW) | Max Monthly Elec (KW) | Electricity Cost – Total ($) | **Total Elec and Fuel Cost ($)** |
|---|---|---|---|---|---|---|---|---|---|
| **AHU3** | 116 | 34,000 | 16,800 | 550,000 | 6,500 | 51,000 | 14 | 9,900 | **16,000** |
| **AHU6** | 140 | 79,000 | 20,000 | 745,000 | 8,800 | 98,000 | 21 | 19,000 | **28,000** |
| **AHU4/5** | 428 | 206,000 | 75,000 | 2,000,000 | 28,000 | 280,000 | 50 | 54,000 | **82,000** |

**Zone Level Optimization (Day and Night)**
**350 annual MTCO2e Saved**
**$126k Cost Savings**

# 41CS Energy Cost Model (ECM3)

**ECM 3:  Resolve Unknown Pressure drop**

- AHU3: Decrease modeled pressure drop due to filters (2026 fan power > 2020 fan power for the same CFM, indicating an additional pressure drop has occurred)

# ECM3: Energy and Cost Savings

Energy Rates:
Electric: $0.19/kWh
Natural Gas: $11.86/mmbtu

| | Mechanical Cooling Load (Tons) | Fan Motor (kWh) | Cooling Elec Energy (CHW or DX) (kWh) | Fuel Input to Heating Plant (kbtu) | Fuel Cost – Total($) | Electrical Energy – Total (kW) | Max Monthly Elec (KW) | Electricity Cost – Total ($) | **Total Elec and Fuel Cost ($)** |
|---|---|---|---|---|---|---|---|---|---|
| **AHU3** | 0 | 72,000 | 150 | 0 | 0 | 72,000 | 9 | 14,000 | **14,000** |

**Resolve Unknown Pressure Drop**
**21 annual MTCO2e Saved**
**$14k Cost Savings**

# 41CS ECM Results

**Cooper Union Carbon Action Plan - ECM Summary**

| Identification | | | | Baseline (Annual) | | | | Proposed (Annual) | | | | Savings (Annual) | | | | Capital and Payback (ROM) | | | | Notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Building | ECM | ECM Description | Model Includes | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Annual Electric | Annual Natural Gas | Annual Energy Cost | Carbon | Capital Expense (ROM) | Potential Rebate (ROM) | After Rebate Expense (ROM) | Simple Payback | |
| | | | | *kWh* | *Therms* | *$* | *MTCO2e* | *kWh* | *Therms* | *$* | *MTCO2e* | *kWh* | *Therms* | *$* | *MTCO2e* | *$* | *$* | *$* | *Years* | |
| 41CS | ECM-1A | Night Setback (AHU-3/6 Off):<br>AHU-3/6 off at night (10 PM to 7 AM) | AHU-3/6 | 731586 | 63966 | 217791 | 551 | 461729 | 39407 | 136312 | 343 | 269,857 | 24,559 | $81,479 | 208.4 | $2,000 | - | $2,000 | 0.1 yrs | BMS schedule change. AHU-3 and AHU-6 off at night. |
| 41CS | ECM-1B | Night Setback (Aggressive):<br>AHU-3/6 reduce by 50% at night<br>AHU-4/5 reduce by 24% CFM at night | AHU-3/4/5/6 | 1750155 | 145795 | 512443 | 1280 | 1540658 | 133190 | 456850 | 1153 | 209,497 | 12,605 | $55,592 | 127.5 | $2,000 | - | $2,000 | - | Aggressive nighttime CFM reduction on AHU-3/4/5/6. Programming only. |
| 41CS | ECM-1C | Night Setback (Moderate):<br>AHU-3/6 reduce by 20% at night<br>AHU-4/5 reduce by 12% CFM at night | AHU-3/4/5/6 | 1750155 | 145795 | 512443 | 1280 | 1651463 | 141166 | 464883 | 1227 | 98,692 | 4,629 | $47,560 | 53.1 | $2,000 | - | $2,000 | - | Moderate nighttime CFM reduction on AHU-3/4/5/6. Programming only. |
| 41CS | ECM-2 | Zone Level Optimization:<br>AHU-3/6: 3,000 to 5,000 CFM reductions<br>AHU-4/5: 13,000 CFM reduction | AHU-3/4/5/6 | 1720485 | 145041 | 505793 | 1267 | 1289782 | 108645 | 379066 | 950 | 430,704 | 36,396 | $126,727 | 317.8 | $30,000 | - | $30,000 | - | Zone-level controls repair (failed valves, dampers, sensors) and setpoint optimization. |
| 41CS | ECM-3 | Resolve AHU-XX Pressure Drop | AHU-3 | 363531 | 34197 | 111082 | 287 | 291437 | 34197 | 97096 | 266 | 72,094 | - | $13,986 | 20.8 | $2,000 | - | $2,000 | - | Investigate root cause of pressure drop. Scope and cost TBD. |

**Implementing all ECMs* can potentially save**

841,000 kWh of Electric

67,000 therms of Natural Gas

$242,000 energy cost

600 annual MTCO2e

*This is for ECM 1a, 1c of only AHU-4/5, 2, and 3. However, more potential savings are possible as ECM 1c is an airflow setback of the current CFM instead of an airflow setback of the optimized CFM from ECM2

# Recommended Future Work for AHUs

- Trouble shoot zones and ensure controls operate as intended
- Move towards implementing aggressive setbacks (including zone level snorkel, fume hood, sash optimization)
- Do field investigation on AHU-3 to resolve pressure drop
- Address AHU-6 flow station limitation to get a more accurate reflection of supply flow



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Thank you!